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# New Late Cretaceous paleomagnetic and geochronologic results from the southern Qiangtang Terrane: Contributions to the reliability of redbed paleomagnetic datasets from the Tibetan Plateau and shape of the Qiangtang Terrane

**Jiahui Ma[1,2], Tianshui Yang[1,2,†], Wenxiao Peng[1,2], Weiwei Bian[1], Suo Wang[1,2], Fei Han[1,2], Jiacheng Liang[1,2], Xianwei Jiao[1,2], Jingjie Jin[2], Shihong Zhang[1,2], Huaichun Wu[1], Haiyan Li[1], and Yiming Ma[3]**

[1]*State Key Laboratory of Biogeology and Environmental Geology, China University of Geosciences, Beijing 100083, China*
[2]*School of Earth Sciences and Resources, China University of Geosciences, Beijing 100083, China*
[3]*School of Earth Sciences, China University of Geosciences, Wuhan 430074, China*

## ABSTRACT

**To better clarify the reliability of redbed paleomagnetic datasets from the Tibetan Plateau and shape of the Qiangtang Terrane (QT) prior to the India-Asia collision, a combined paleomagnetic and detrital zircon U-Pb geochronologic study is conducted on the Upper Cretaceous Abushan Formation (Fm) redbeds, dated to be within 89.1–83.6 Ma, in the Anduo area. The tilt-corrected grand mean direction for 37 sites is $D_s = 0.9°$, $I_s = +36.4°$, $k = 43.6$, $\alpha_{95} = 3.6°$, which provides a paleopole at 78.2°N, 266.8°E ($A_{95} = 3.6°$), corresponding to a paleolatitude of 20.5° ± 3.6°N for the study area (32.3°N, 91.4°E). Reliable demagnetization curves, positive fold, and conglomerate tests support the interpretation that characteristic remanent magnetization directions recorded primary magnetizations carried by detrital hematite and did not suffer from the influence of distortional strains. Our paleomagnetic results indicate that the mean inclination observed from the southwest dipping limb ($I_s = +44.2°$) is clearly steeper than that from the north dipping limb ($I_s = +34.2°$). The results of syntectonic-sedimentation-correction and the fluvial gravel deposits present in the Abushan Fm redbeds sampled support the contention that the apparent inclination discrepancy is attributed to the syntectonic growth strata and that the paleomagnetic datasets observed from two limbs of folds are still reliable. Furthermore, reliable Late Cretaceous paleomagnetic datasets show that the shape of the QT west of the Eastern Himalaya Syntaxis (EHS) during the Late Cretaceous was similar to its present-day relatively E-W alignment, and that around the EHS, its shape changed from the Late Cretaceous NE-SW alignment to the present-day approximately NW-SE alignment.**

## INTRODUCTION

The Tibetan Plateau is regarded as a complex tectonic collage of several microcontinents, flysch complexes, and island arcs that have gradually amalgamated with paleo-Asia since the early Paleozoic (e.g., Yin and Harrison, 2000). The Qiangtang, Lhasa, and Himalayan terranes are traditionally considered to have drifted northward from Gondwana in the south, and their eventual paleo-Asian amalgamations are along the Jinsha suture zone, Bangong-Nujiang suture zone, and Indus-Tsangpo suture zone, respectively (e.g., Yin and Harrison, 2000; Pan et al., 2012; Metcalfe, 2013; Kapp and DeCelles, 2019; Fig. 1A). Notably, the early Cenozoic India-Asia collision led to the closure of the Neo-Tethyan Ocean, and the subsequent continued plate convergence resulted in the extensive intracontinental deformation within Asia and the uplift of the Tibetan Plateau that changed the Asian climatic system (e.g., Yin and Harrison, 2000; van Hinsbergen et al., 2019). Therefore, determining the location of the southern margin of paleo-Asia during the late Mesozoic and estimating the intracontinental shortening amount within Asia after the early Cenozoic India-Asia collision are vital to better understanding the formation and evolution of the Tibetan Plateau and the Asian climatic system (e.g., van Hinsbergen and Boschman, 2019).

The late Mesozoic Lhasa-Qiangtang and early Cenozoic India-Asia collisions together with subsequent penetration of the Indian plate into Asia generally occurred in a north-south direction along the Bangong-Nujiang suture zone and Indus-Tsangpo suture zone. Paleomagnetic is one of the major techniques that can effectively quantify plate paleolatitude to estimate kinematic plate movements. Therefore, many paleomagnetic studies of Cretaceous rocks have been performed in the Lhasa and Qiangtang terranes (e.g., Pozzi et al., 1982; Lin and Watts, 1988; Sun et al., 2012; Chen et al., 2012, 2017; Ma et al., 2014, 2017b, 2018, 2022; Yang et al., 2015; Bian et al., 2017, 2020, 2022a; Cao et al., 2017a, 2017b, 2020; Li et al., 2017b; Wang et al., 2022). The paleomagnetic data from the coeval rocks yielded significantly different inclinations and thus paleolatitudes estimates. For example, the Late Cretaceous paleomagnetic data reported from both volcanic and sedimentary rocks of the Lhasa terrane produced significantly different inclinations, ranging from 16.2° to 48.5°, with the corresponding paleolatitudes ranging from 6.6°N to 25.4°N for a reference latitude of 29.3°N (e.g., Bian et al., 2022a). These discrepant paleolatitude estimates resulted in a range of India-Asia collision ages varying from 65 Ma to 43 Ma, as well as the significantly discrepant shortening magnitudes, which have been estimated to be from 1900 km (e.g., Tang et al., 2013) to only a few hundred kilometers (e.g., Tan et al., 2010) between the Lhasa terrane and stable Asia.

Because paleomagnetic records from individual volcanic units provide only spot readings

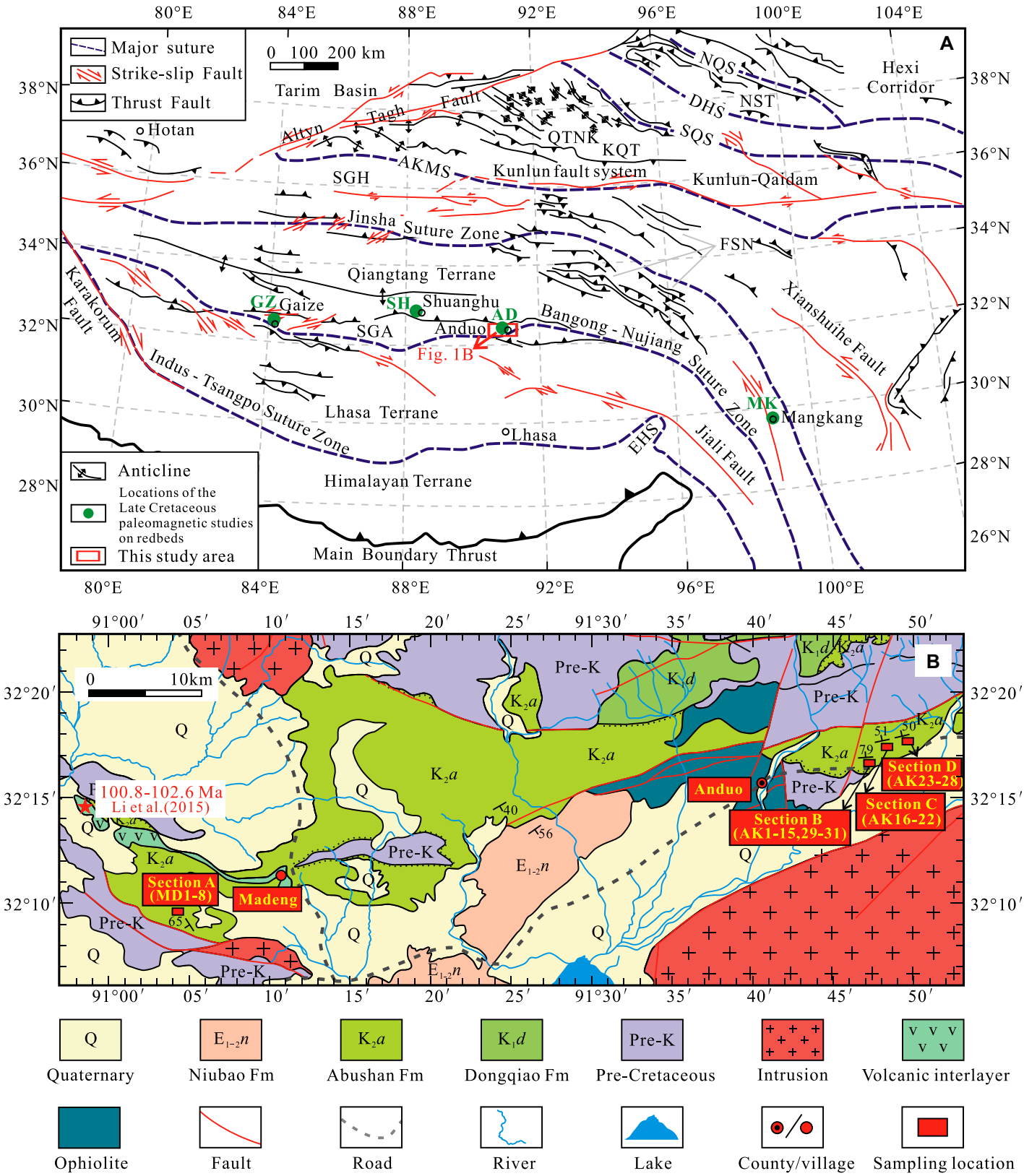
Tianshui Yang https://orcid.org/0000-0001-5530-0025
†yangtsh@cugb.edu.cn

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of the geomagnetic field behavior and their reliability totally depends on the sampled time length and number of sites, some of the paleomagnetic results observed from the Late Cretaceous volcanic rocks could be attributed to the spot readings of the geomagnetic field (e.g., Sun et al., 2012). Considering that compaction-induced inclination shallowing probably affected data from sedimentary rocks (e.g.,



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**Figure 1. Generalized geologic map showing the paleomagnetic sampling area for this study. (A) Tectonic map of the Tibetan Plateau and surrounding areas, modified after Yang et al. (2015), showing the locations of previous studies that sampled Upper Cretaceous redbeds of the Qiangtang Terrane. AD—sampling location of this study; AKMS—Ayimaqin-Kunlun-Muztagh suture; DHS—Danghe Nan Shan suture; EHS—Eastern Himalaya Syntaxis; FSN—Fenghuo Shan-Nangqian fold and thrust belt; GZ—sampling location of Chen et al. (2017); KQT—Kunlun-Qaidam terrane; MK—sampling location of Huang et al. (1992) and Tong et al. (2015); NST—Nan Shan thrust belt; QTNK—Qimen Tagh-North Kunlun thrust system; NQS—north Qilian Suture; SGA—Shiquanhe-Gaize-Anduo thrust system; SGH—Songpan-Ganzi-Hoh Xil terrane; SH—sampling location of Meng et al. (2018); SQS—South Qilian suture. (B) Simplified geologic map of the sampling area in the Anduo area showing sampling locations. Fm—Formation.**

Gilder et al., 2001; Tan et al., 2003; Yan et al., 2005; Kodama, 2012; Huang et al., 2013; Ding et al., 2015), some researchers suggested that the lower paleolatitudes obtained from Cretaceous sedimentary rocks could result from inclination shallowing (e.g., Tan et al., 2010; Chen et al., 2017). For example, Chen et al. (2017) carried out a combined paleomagnetic and geochronologic investigation on Cretaceous rocks in the Gaize area of the southern Qiangtang Terrane (QT). The mean inclination of 22 sites from the Upper Cretaceous redbeds is 13.1° shallower than that based on data from 14 sites from the underlying ca. 110–100 Ma volcanic rocks. In contrast, the mean inclination of ~47.9° after the elongation/inclination (E/I) inclination shallowing correction was consistent with the ~47.7° result from the volcanic rocks (Tauxe and Kent, 2004). Furthermore, the Late Cretaceous paleomagnetic data from the Linzhou Basin of the Lhasa terrane (Tan et al., 2010) also revealed the similar result, which indicated that the tilt-corrected mean inclination (23.5° ± 2.5°) of 43 redbeds sites was significantly shallower than that (41.9° ± 4.4°) of 32 lava flows sites and the mean inclination (~42.0°) of the redbeds after the elongation/inclination (E/I) shallowing correction was similar with the characteristic remanent magnetization (ChRM) mean inclination of the lava flows. Tan et al. (2010) proposed that the Upper Cretaceous redbeds of the Lhasa terrane experienced ~18° of inclination flattening. However, such an observation is challenged by the fact that no significant inclination discrepancy has been recognized between the data from redbeds and intercalated volcanic rocks in the Tibetan Plateau (Sun et al., 2006; Cao et al., 2017a). The Late Cretaceous paleomagnetic result reported by Cao et al. (2017a) from the Linzhou Basin of the Lhasa terrane revealed that the tilt-corrected mean direction ($D_s = 1.7°$, $I_s = +17.9°$, $k_s = 38.1$, $\alpha_{95} = 7.9°$, $n = 10$) for the Shexing Formation (Fm) redbeds was consistent with the one ($D_s = 359.4°$, $I_s = +22.5°$, $k_s = 18.9$, $\alpha_{95} = 10.8°$, $n = 11$) for the intercalated basalt flows. Another paleomagnetic study on the Early Cretaceous basalt flows intercalated with the redbeds from the Qaidam Basin of the northeastern Tibetan Plateau also indicated that the tilt-corrected site-mean direction for the redbeds ($D_s = 18.4°$, $I_s = +55.3°$, $k_s = 69.1$, $\alpha_{95} = 5.5°$, $n = 11$) and the basalt flows ($D_s = 16.1°$, $I_s = +52.0°$, $k_s = 42.6$, $\alpha_{95} = 8.6°$, $n = 8$) are entirely consistent (Sun et al., 2006). Furthermore, Yang et al. (2019) summarized the available Cretaceous paleomagnetic datasets from the Lhasa terrane. They showed that for the same reference point (29.3°N, 85.3°E) the data from 91 sites in Late Cretaceous volcanic rocks yielded a paleolatitude of 14.0° ± 2.1°N, which is very consistent with 13.5° ± 1.2°N deduced from the 118 reliable Late Cretaceous sedimentary sites. Therefore, high-quality paleomagnetic datasets are still needed to ascertain if the compaction-induced inclination shallowing is substantial in the Cretaceous redbeds of the Tibetan Plateau.

In addition to the deposition and/or compaction-induced inclination shallowing, some researchers considered the possibility that unidentified syntectonic growth strata originally tilted to the north or south could lead to significant inclination deviations if the sampling strata were not restored to their originally nonhorizontal orientation at the time of remanence acquisition (e.g., Gilder et al., 2001; Tong et al., 2019; Bian et al., 2020; Bian and Yang, 2021; Wang et al., 2022). For example, after analyzing the Cretaceous paleomagnetic data from the Lhasa terrane, Tong et al. (2019) found that paleomagnetic data obtained from the southern limb of folds yielded steeper site-mean inclinations than those from the northern limb of folds, and proposed that the significant inclination discrepancy obtained from two limbs of folds could result from the unidentified syntectonic growth strata, which were not restored to their original orientation before folding. Bian et al. (2020) performed a paleomagnetic study on the Upper Cretaceous Jingzhushan Fm redbeds of the western Lhasa terrane. Though the mean direction of all the 54 paleomagnetic sites passed positive fold tests, the mean inclination ($I_s = +51.8°$) obtained from the southern limb of folds is clearly steeper than that ($I_s = +19.0°$) observed from the northern limb of folds. After analyzing several possible factors, Bian et al. (2020) concluded that the ChRM direction observed from the Jingzhushan Fm redbeds should represent primary magnetization and the large inclination discrepancy of the two limbs of folds could be attributed to the syntectonic-sedimentation. Because the steeper inclinations obtained from the southern limb of folds can reasonably offset the shallower inclinations obtained from the northern limb of folds, some researchers proposed that paleomagnetic data obtained from two limbs of the folded syntectonic strata are still reliable for paleogeographic reconstructions (e.g., Tong et al., 2019; Bian et al., 2020; Bian and Yang, 2021; Wang et al., 2022). However, Zhao et al. (2021) argued that the Upper Cretaceous Jingzhushan Fm redbeds in the northern Lhasa terrane suffered from post-depositional tectonic penetrative deformation, and thus considered that the tectonic strains led to the steeper inclinations found in the southern limb of folds and the shallower inclinations of the northern limb of folds.

It should be emphasized that sedimentary rocks, especially redbeds, have always been the most widely used material for paleomagnetic studies in the Tibetan Plateau and its adjacent areas due to their widespread distribution (e.g., Achache et al., 1984; Sun et al., 2006, 2012; Ding et al., 2014; Yang et al., 2015, 2019; Yan et al., 2016; Chen et al., 2017; Tong et al., 2017, 2022; Bian et al., 2020, 2022a, 2022b; Cao et al., 2020; Li et al., 2020a, 2020b; Yuan et al., 2021; Wang et al., 2022; Ma et al., 2022), as well as their ability to record nearly continuous geomagnetic fields and, usually, unequivocal paleohorizontal, which are occasionally the problem with volcanic rocks. Due to the India-Asia collision and subsequent continued convergence, post-depositional tectonic strains generally have affected the pre-Paleogene strata exposed on the Tibetan Plateau. Therefore, it is especially important to clarify the reliability of ChRM directions recorded in the redbeds of the Tibetan Plateau.

In this study, we conducted a paleomagnetic study on the Upper Cretaceous Abushan Fm redbeds in the Anduo area of the QT. Our new paleomagnetic datasets are obtained from two different limbs of the folded Abushan Fm redbeds and provide the opportunity for the positive fold and conglomerate tests that are essential to evaluate the reliability of paleomagnetic data (e.g., Meert et al., 2020; Heslop



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and Roberts, 2018). We included tests such as the paleosecular variation (Deenen et al., 2011), the origin of hematite (e.g., Jiang et al., 2015, 2022), the anisotropy of magnetic susceptibility (AMS) analysis related to the distortional strains (e.g., Li et al., 2010; Parés, 2015; Hrouda and Chadima, 2020; Cao et al., 2022), the inclination shallowing correction (e.g., Hodych and Buchan, 1994; Tauxe and Kent, 2004), and the syntectonic-sedimentation-correction (Tong et al., 2019). Therefore, these new data can be utilized to address the reliability of the redbed paleomagnetic datasets from the Tibetan Plateau and the cause of their inclination discrepancies, as well as constraining the shape of the QT prior to the India-Asia collision.

## GEOLOGY AND SAMPLING

The QT is located in the central Tibetan Plateau, and it is separated from the Lhasa terrane by the Bangong-Nujiang suture zone on the south and from the Songpan-Ganzi terrane by the Jinsha suture zone on the north (Fig. 1A). The present-day QT extends from west to east and rotates to the N-S trend after passing through the Eastern Himalaya Syntaxis (EHS) (Tapponnier et al., 1982; Armijo et al., 1986; Royden et al., 1997; Wang and Burchfiel, 2000; Jessup et al., 2008). Our study area is located in the southern margin of the QT where the Abushan Fm redbeds are widely exposed (Fig. 1B). Based on the 1:250,000 scale Anduo (I46C004001, 2003) and Zigetangcuo (I46C004002, 2005) regional geological survey reports, the middle-upper part of the Abushan Fm includes purplish-red sandstones and siltstones with a fining-upward trend, and its lower part mainly consists of gray-red conglomerates interbedded with sandstones. In the Anduo area, the Abushan Fm is ~750 m thick, and unconformably overlies the Upper Jurassic–Lower Cretaceous Bangai Fm and underlies the Eocene Niubao Fm (I46C004001, 2003; Fig. 2). In the Madeng area, this formation is ~500 m thick with volcanic interlayers in the lower part, and unconformably overlies the Upper Jurassic Suowa Fm and underlies the Eocene Niubao Fm (I46C004002, 2005; Fig. 2). Two volcanic interlayers located at the bottom of the Abushan Fm

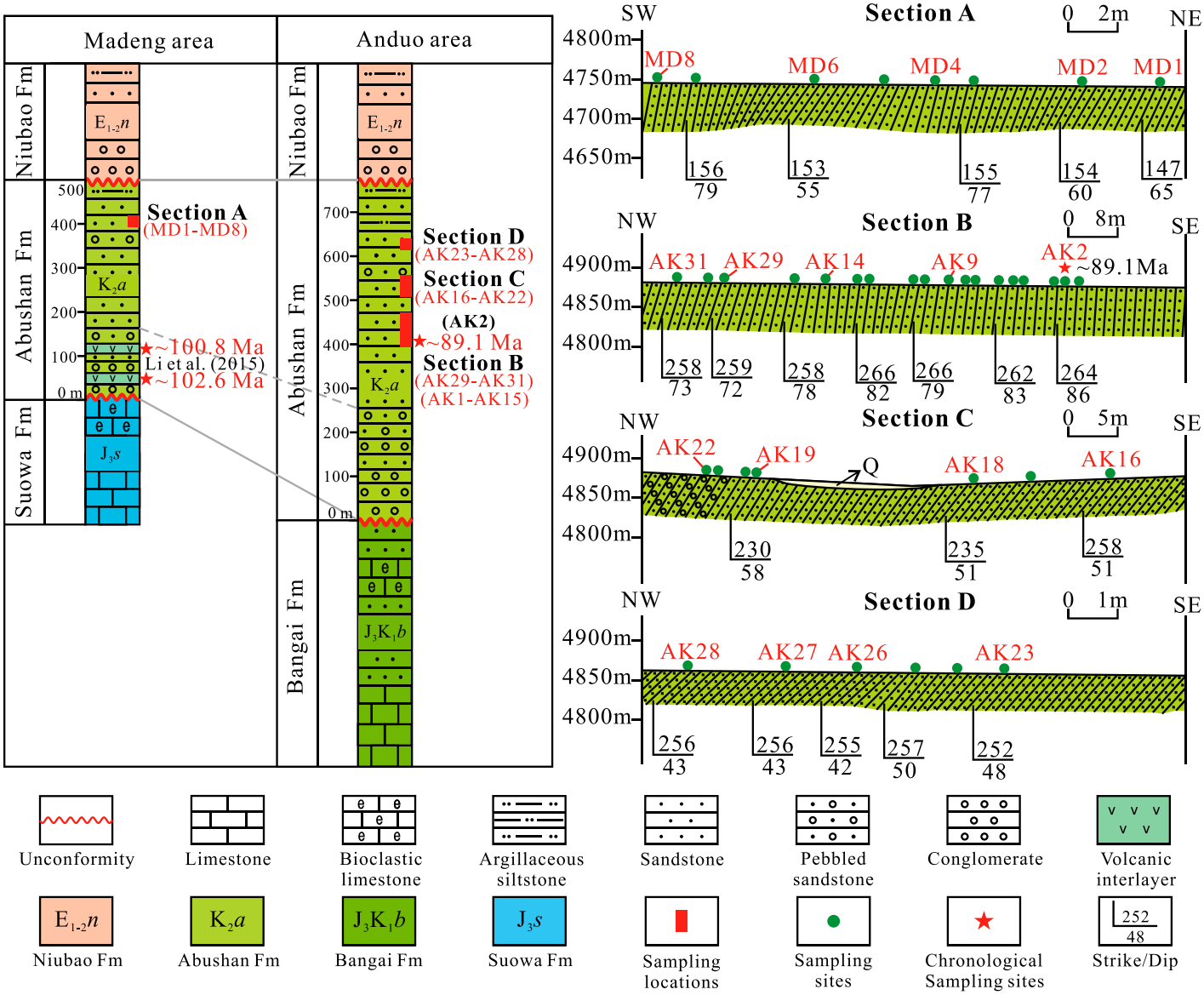



**Figure 2. Stratigraphic columns and cross sections for the field sampling of the Upper Cretaceous Abushan Formation (Fm) redbeds of the Qiangtang Terrane, Tibetan Plateau.**



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in the Madeng area yielded zircon U-Pb ages of 100.8 ± 0.9 Ma and 102.6 ± 1.6 Ma (Li et al., 2015; Fig. 2). Based on the above age constraints and considering that our new youngest detrital zircon U-Pb dating is 89.1 ± 0.9 Ma and all the Abushan Fm specimens produce normal ChRM directions, as well as the long Cretaceous Normal Polarity Superchron, which spans a time interval of ca. 121.4–83.6 Ma (Ogg, 2020), the depositional ages of the sampled Abushan Fm redbeds can be restricted within 89.1–83.6 Ma. The oldest phase of folding of the Late Cretaceous strata occurred in the latest Late Cretaceous (I46C004002, 2005).

A total of 39 paleomagnetic sites were collected from the middle-upper of the Abushan Fm redbeds (Figs. 1B and 2). Eight sites (MD1–MD8) were collected from Section A (32°13′47″N, 91°02′46″E), located ~5 km southwest of the Madeng village, where the bedding dips from 55° to 79° toward the southwest (Figs. 2 and 3). Thirty-one sites were drilled from Sections B (AK1-AK15 and AK29-AK31, 32°16′34″N, 91°47′27″E), C (AK16-AK22, 32°17′17″N, 91°49′22″E), and D (AK23-AK28, 32°17′94″N, 91°48′28″E), located ~15 km east of the Anduo county, where the bedding dips from 42° to 84° toward the north (Figs. 2 and 3). To perform a conglomerate test and assure the reliability of the ChRM direction, two sites (AK21 and AK22) including 20 oriented cores were drilled from redbed cobbles of conglomerate layers that are intercalated with the Upper Cretaceous Abushan Fm redbeds. The bedding attitudes for the 39 paleomagnetic sites are given in Table 1. Generally, each paleomagnetic site includes ~10 cores, and each oriented core of the same site was distributed at different stratigraphic levels (Fig. 3). All cores were collected with a gasoline-powered drill and oriented by both a sun compass and a magnetic compass. The discrepancy between these two oriented records was less than 3°, indicating that the local magnetic disturbance can be ignored.

Eight hand blocks oriented on the strata bedding plane were collected from Section B for testing the anisotropy-based inclination shallowing. Furthermore, one fresh block sample for detrital zircon U-Pb geochronology was collected from the AK2 site, which is located at the bottom of the sampled Abushan Fm redbeds sequence.

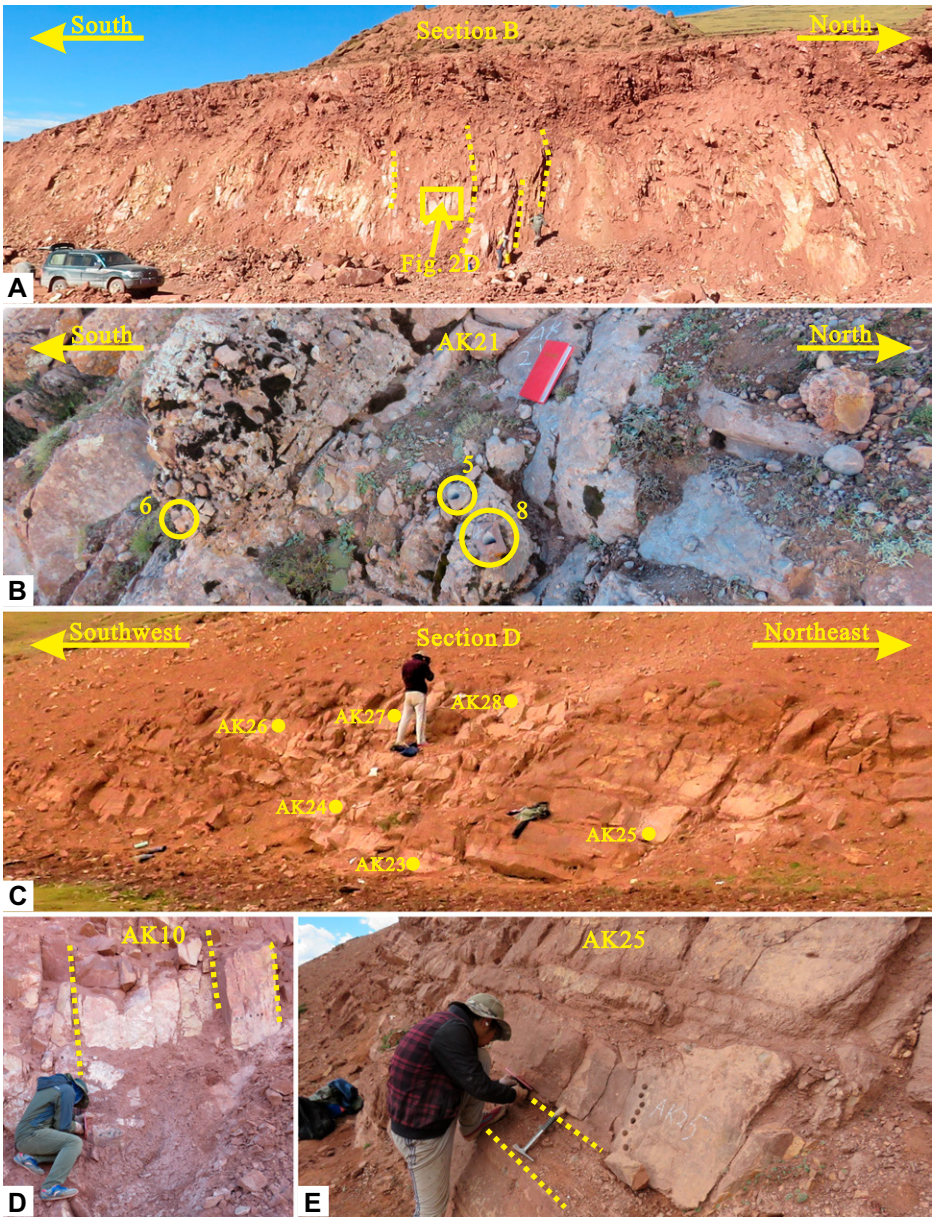



**Figure 3. Field photographs showing exposures of the Upper Cretaceous Abushan Formation in the southern Qiangtang Terrane, Tibetan Plateau.**

## METHODS

The standard oriented cores were cut into 2.2-cm-long cylindrical specimens for subsequent analysis. All redbeds specimens were subjected to stepwise thermal demagnetization up to 680 °C using an MMTDSC or an ASC-TD 48 thermal demagnetizer. Remanent magnetization was measured using the 2G-755-4 K cryogenic magnetometer. The isothermal remanent magnetization (IRM), backfield demagnetization of the saturation IRM (SIRM), and three-axis IRM were performed using an ASC IM10-30 pulse magnetizer and measured with a JR-6A spinner magnetometer. Three-axis IRM were imparted in successively smaller fields of 2.4 T, 0.4 T, and 0.12 T along three orthogonal directions (Lowrie, 1990). AMS analysis was carried out using an MFK1 Kappabridge (AGICO) before thermal demagnetization.

All demagnetization and remanence measurements were performed within the magnetically shielded room (<300 nT) at the Paleomagnetic and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing, China. ChRM directions were determined using principal component analysis (Kirschvink, 1980), and site-mean directions were calculated using Fisherian statistics (Fisher, 1953). Paleomagnetic data were processed using the packages developed by Enkin (1990) and Cogné (2003).

A total of eight standard core samples were drilled in the direction perpendicular to the strata bedding from the oriented hand blocks. These standard specimens were subjected to successively increasing DC fields in the range of



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TABLE 1. SITE MEAN DIRECTIONS OF THE ABUSHAN FORMATION REDBEDS FROM THE ANDUO AREA IN THE QIANGTANG TERRANE, TIBETAN PLATEAU

| Site | Strike/dip (°) | n/N | In situ | | Tilt-corrected | | After correction | | $k$ | $\alpha_{95}$ (°) | VGP | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $D_g$ (°) | $I_g$ (°) | $D_s$ (°) | $I_s$ (°) | $D_c$ (°) | $I_c$ (°) | | | Plon (°E) | Plat (°N) |
| Section A (32°13′47″N, 91°02′46″E) | | | | | | | | | | | | |
| MD1 | 147/65 | 7/8 | 25.8 | −13.0 | 14.8 | 41.4 | 18.7 | 35.7 | 18.4 | 14.5 | 210.8 | 74.4 |
| MD2 | 154/60 | 7/8 | 21.8 | 8.8 | 353.4 | 45.3 | 0.1 | 42.4 | 22.0 | 13.1 | 319.3 | 82.1 |
| MD3 | 155/77 | 10/10 | 30.3 | −9.5 | 6.4 | 48.8 | 12.9 | 44.5 | 67.6 | 5.9 | 204.8 | 83.9 |
| MD4 | 155/77 | 10/10 | 16.4 | −1.2 | 347.7 | 39.8 | 353.5 | 37.8 | 37.3 | 8.0 | 322.8 | 75.4 |
| MD5 | 153/55 | 8/10 | 21.1 | 12.5 | 353.2 | 46.1 | 360.0 | 43.1 | 79.1 | 6.3 | 323.7 | 82.4 |
| MD6 | 153/55 | 9/10 | 21.3 | 10.7 | 355.3 | 45.0 | 1.7 | 41.8 | 73.9 | 6.0 | 308.1 | 83.0 |
| MD7 | 156/79 | 7/8 | 17.8 | 5.6 | 338.4 | 42.0 | 345.0 | 41.2 | 54.5 | 8.2 | 343.9 | 69.4 |
| MD8 | 156/79 | 6/8 | 17.5 | 0.7 | 344.8 | 40.8 | 351.0 | 39.3 | 50.3 | 9.5 | 331.5 | 73.9 |
| Overall mean A | | | 21.5 | 1.8 | 354.0 | 44.2 | 0.4 | 41.2 | 81.7 | 6.2 | 311.8 | 81.9 |
| N = 8 sites | | | | | | $k_c$ = 81.7 | | $\alpha_{95c}$ = 6.2 | | | $A_{95}$ = 7.5 | |
| Section B (32°16′34″N, 91°47′27″E) | | | | | | | | | | | | |
| AK1 | 264/86 | 6/8 | 160.1 | 42.1 | 10.0 | 49.9 | 13.1 | 57.1 | 28.0 | 12.9 | 189.4 | 81.3 |
| AK2 | 264/86 | 5/8 | 147.5 | 44.2 | 20.2 | 43.5 | 23.9 | 50.1 | 37.6 | 12.6 | 197.6 | 71.0 |
| AK3 | 264/86 | 7/8 | 146.1 | 44.6 | 20.9 | 42.5 | 24.6 | 49.2 | 17.1 | 15.0 | 198.8 | 70.1 |
| AK4 | 262/83 | 7/8 | 124.9 | 66.4 | 10.5 | 22.4 | 11.7 | 29.6 | 49.7 | 8.6 | 244.0 | 67.2 |
| AK5 | 262/83 | 10/10 | 142.5 | 59.6 | 9.2 | 32.8 | 11.0 | 40.0 | 37.8 | 8.0 | 239.3 | 73.3 |
| AK6 | 262/83 | 6/8 | 157.3 | 59.0 | 1.4 | 36.8 | 2.5 | 44.2 | 29.1 | 12.6 | 265.0 | 78.1 |
| AK7 | 263/82 | 7/8 | 132.1 | 65.6 | 10.5 | 25.8 | 11.8 | 33.0 | 56.9 | 8.1 | 241.8 | 69.0 |
| AK8 | 263/82 | 10/10 | 128.8 | 66.6 | 10.7 | 24.2 | 11.9 | 31.3 | 76.6 | 5.6 | 242.4 | 68.1 |
| AK9 | 263/82 | 7/8 | 150.1 | 67.4 | 2.7 | 28.6 | 3.7 | 36.0 | 29.5 | 11.3 | 262.6 | 72.8 |
| AK10 | 266/79 | 8/8 | 140.1 | 61.9 | 15.2 | 32.8 | 17.1 | 39.9 | 37.7 | 9.1 | 224.3 | 70.1 |
| AK11 | 266/79 | 10/10 | 141.0 | 63.8 | 13.3 | 31.8 | 15.0 | 38.9 | 89.5 | 5.1 | 229.7 | 70.7 |
| AK12 | 266/82 | 6/8 | 145.2 | 63.1 | 11.6 | 30.6 | 13.0 | 37.8 | 36.7 | 11.2 | 235.1 | 71.0 |
| AK13 | 266/82 | 7/8 | 153.8 | 50.4 | 15.5 | 43.8 | 18.4 | 50.8 | 42.0 | 9.4 | 203.7 | 74.9 |
| AK14 | 255/84 | 5/8 | 148.3 | 54.4 | 357.6 | 39.8 | 359.2 | 47.1 | 104.8 | 7.5 | 284.4 | 80.1 |
| AK15 | 258/78 | 5/8 | 138.2 | 62.0 | 4.7 | 35.6 | 6.5 | 42.7 | 33.5 | 13.4 | 251.8 | 76.7 |
| AK29 | 259/72 | 5/8 | 132.6 | 63.6 | 8.6 | 38.1 | 11.0 | 45.1 | 30.6 | 14.0 | 234.2 | 76.7 |
| AK30 | 258/73 | 7/8 | 136.8 | 61.2 | 8.9 | 41.1 | 9.7 | 47.9 | 37.3 | 10.0 | 227.2 | 78.3 |
| AK31 | 258/73 | 6/8 | 117.7 | 58.1 | 17.7 | 34.8 | 20.7 | 41.2 | 39.5 | 10.8 | 216.7 | 69.4 |
| Section C (32°17′17″N, 91°49′22″E) | | | | | | | | | | | | |
| AK16 | 258/51 | 8/8 | 54.0 | 73.6 | 348.4 | 36.0 | 7.1 | 38.1 | 63.2 | 7.0 | 314.3 | 73.9 |
| AK17 | 258/51 | 5/8 | 30.3 | 71.8 | 346.4 | 29.0 | 2.4 | 32.0 | 48.8 | 11.1 | 310.9 | 69.1 |
| AK18 | 235/51 | 7/9 | 11.8 | 64.0 | 344.8 | 19.4 | 347.9 | 26.4 | 64.0 | 7.6 | 307.0 | 63.6 |
| AK19 | 230/58 | 7/9 | 44.5 | 67.8 | 345.8 | 27.9 | 347.0 | 34.1 | 48.8 | 8.7 | 311.2 | 68.3 |
| AK20 | 230/58 | 8/8 | 55.3 | 65.7 | 348.6 | 31.0 | 351.1 | 37.5 | 46.6 | 8.2 | 307.6 | 71.3 |
| Section D (32°17′94″N, 91°48′28″E) | | | | | | | | | | | | |
| AK23 | 252/48 | 5/8 | 1.9 | 77.0 | 345.5 | 28.8 | 347.5 | 37.1 | 131.7 | 6.7 | 312.0 | 68.1 |
| AK24 | 252/48 | 5/8 | 9.4 | 76.8 | 348.9 | 30.1 | 349.6 | 37.5 | 100.8 | 7.7 | 306.0 | 71.0 |
| AK25 | 257/50 | 7/9 | 34.0 | 72.8 | 1.1 | 27.4 | 2.2 | 34.6 | 78.3 | 6.9 | 267.9 | 72.2 |
| AK26 | 255/42 | 9/10 | 39.5 | 75.1 | 0.5 | 38.2 | 2.3 | 45.4 | 67.2 | 6.3 | 268.9 | 79.2 |
| AK27 | 256/43 | 6/8 | 14.4 | 82.8 | 350.5 | 40.5 | 351.1 | 48.0 | 86.9 | 7.2 | 316.2 | 77.6 |
| AK28 | 256/43 | 7/8 | 12.7 | 73.7 | 354.5 | 32.0 | 355.4 | 39.5 | 91.2 | 6.4 | 291.1 | 74.2 |
| Overall mean B | | | 122.9 | 71.8 | 2.6 | 34.2 | 5.3 | 40.9 | 46.4 | 4.0 | 260.1 | 76.4 |
| N = 29 sites | | | | | | $k_c$ = 53.3 | | $\alpha_{95c}$ = 3.7 | | | $A_{95}$ = 4.0 | |
| Overall mean C | | | 74.6 | 68.3 | 0.9 | 36.4 | 4.2 | 41.0 | 43.6 | 3.6 | 266.8 | 78.2 |
| N = 37 sites | | | | | | $k_c$ = 57.7 | | $\alpha_{95c}$ = 3.1 | | | $A_{95}$ = 3.6 | |

*Note*: n/N—number of samples used to calculate mean and measured; $D_g/I_g$ ($D_s/I_s$)—declination and inclination in geographic (stratigraphic) coordinates; $D_c/I_c$—declination and inclination after syntectonic-sedimentation-correction; $k$—the best estimate of the precision parameter; $\alpha_{95}$ ($A_{95}$)—the radius that the mean direction (pole) lies within 95% confidence; $k_c$—the best estimate of the precision parameter after syntectonic-sedimentation-correction; $\alpha_{95c}$—the radius that the mean direction (pole) lies within 95% confidence after syntectonic-sedimentation-correction; VGP—virtual geomagnetic pole; Plat—latitude of the pole; Plon—longitude of pole. (1) Overall mean A (section A, N = 8 sites): The McFadden (1990) fold test is positive at 95% confidence level at "Xi2" text: critical Xi at 95% = 3.30 and at 99% = 4.56, respectively. "Xi2" IS = 3.63, "Xi2" TC = 0.64. (2) Overall mean B (N = 29 sites): The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: $k_s/k_g = 3.25 > F(2*(n2–1), (n1–1))$ at 5% and 1% point = 1.56 and 1.87, respectively. The McFadden (1990) fold test is positive at 95% and 99% confidence levels at "Xi2" text: critical Xi at 95% = 6.26 and at 99% = 8.86, respectively. "Xi2" IS = 25.08, "Xi2" TC = 0.76. (3) Overall mean C (N = 37 sites): The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: $k_s/k_g = 11.00 > F(2*(n2–1), (n1–1))$ at 5% and 1% point = 1.47 and 1.74, respectively. The McFadden (1990) fold test is positive at 99% confidence level at "Xi1" text: critical Xi at 95% = 7.08 and at 99% = 10.00, respectively. "Xi1" IS = 16.37, "Xi1" TC = 9.80; and is positive at 99% confidence level at "Xi2" text: "Xi2" IS = 33.35, "Xi2" TC = 8.21. (4) Overall mean D (N = 37 sites) after syntectonic-sedimentation-correction: The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: $k_s/k_g = 8.19 > F(2*(n2–1), (n1–1))$ at 5% and 1% point = 1.47 and 1.74, respectively. The McFadden (1990) fold test is positive at 95% and 99% confidence levels at "Xi2" text: critical Xi at 95% = 7.08 and at 99% = 10.00, respectively. "Xi2" IS = 33.77, "Xi2" TC = 2.45.

0–1000 mT along the 45° angle to the sampling strata bedding plane. While applicating the DC field, the IRM parallel (IRMx) and perpendicular (IRMz) to the bedding plane were measured with a JR-6A spinner magnetometer. After the IRM acquired in 1000 mT, a stepwise thermal demagnetization was performed (Hodych and Buchan, 1994).

Polished thin sections for microscopic investigation were made from the 2.5-cm-diameter specimens. And the microscopic observations of magnetic carriers were studied using an optical microscope (ECLIPSE LV100N POL) under reflected light.

Detrital zircon grains from the block sample AK2 were separated by standard density and magnetic separation techniques. Cathodoluminescence (CL) images were performed to select the zircon analytical spots without crack and inclusion. The U-Pb zircon dating was carried out using the laser ablation inductively coupled plasma–mass spectrometry (ICP-MS) technique at the Mineral Laser Microprobe Analysis Laboratory of the China University of Geosciences, Beijing. The standard material (NIST612) was chosen as an external standard to calibrate the U-Th-Pb isotope ratio. The method described in Andersen (2002) has been used to correct for common lead. The ICP-MS DataCal (Liu et al., 2010) and Isoplot (Ludwig, 2003) programs were used for age calculations and for plotting concordia diagrams.



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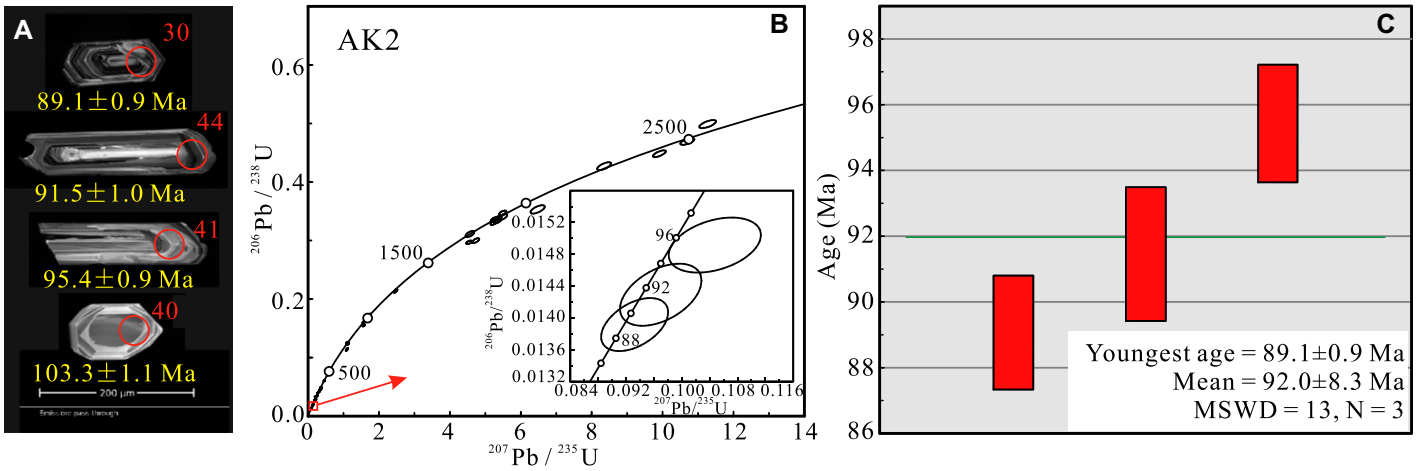



**Figure 4. Detrital zircon U-Pb dating of samples from the Qiangtang Terrane, Tibetan Plateau. (A) Cathodoluminescence images of representative detrital zircons and corresponding $^{206}Pb/^{238}U$ ages of the individual analyzed spots. (B) U-Pb concordia diagram for detrital zircons. (C) Bar plot shows the weighted mean $^{206}Pb/^{238}U$ ages. MSWD—mean square of weighted deviates; N—number of analyzed spots.**

## RESULTS

### Detrital Zircon U-Pb Analytical Results

The CL images showed that most of detrital zircons are euhedral to subhedral with grain sizes ranging from 100 μm to 260 μm with clearly oscillatory zoning (Fig. 4A). These features, combined with higher Th/U ratios of all zircons compared to metamorphic zircons, suggest that the dated zircons are igneous in origin. A total of 60 of 63 zircon grains analyzed provided reliable U-Pb ages, 35 of them produced pre-Paleozoic dates, and the remaining 25 grains yielded Mesozoic dates. Considering that three zircon grains exhibited Late Cretaceous dates of 89.1 ± 0.9 Ma, 91.5 ± 1.0 Ma, and 95.4 ± 0.9 Ma (Figs. 4B and 4C), we use the youngest zircon age of 89.1 ± 0.9 Ma to constrain the maximum depositional age of the sampled Abushan Fm redbeds sequence.

### Anisotropy of Magnetic Susceptibility Results

The maximum ($K_{max}$), intermediate ($K_{int}$), and minimum ($K_{min}$) principal axes of the AMS ellipsoids in equal-area projections show the $K_{max}$ axes are in bedding planes and slightly grouped in an approximately northwest-southeast direction (Fig. 5). The $K_{min}$ axes are roughly clustered perpendicular to bedding planes with a slight trend to the southwest. Such a distribution of the principal axes is common in primary sedimentary fabrics with weak tectonic fabric and comparable with those observed in foreland basin sedimentary rocks (e.g., Gilder et al., 2001; Parés, 2004; Huang et al., 2006, 2010; Tang et al., 2013). The corrected anisotropy degree ($Pj < 1.05$) is relatively low, and their shape parameter ($-1 < T < +1$, usually $T > 0$) indicates a range from obviously oblate to prolate (Fig. 5). These AMS characteristics are consistent with the incipient deformation (stage 2) to pencil structure (stage 3) as described by Hrouda and Chadima (2020), indicating that the studied Abushan Fm redbeds did not suffer from strong deformation.

### Rock-Magnetic Results

Rock-magnetic results show that the Abushan Fm redbed specimens have the same rock-magnetic characteristics (Fig. 6). The IRM acquisition curves reveal a gradual increase and the saturation is not reached even at 2.4 T, indicat-

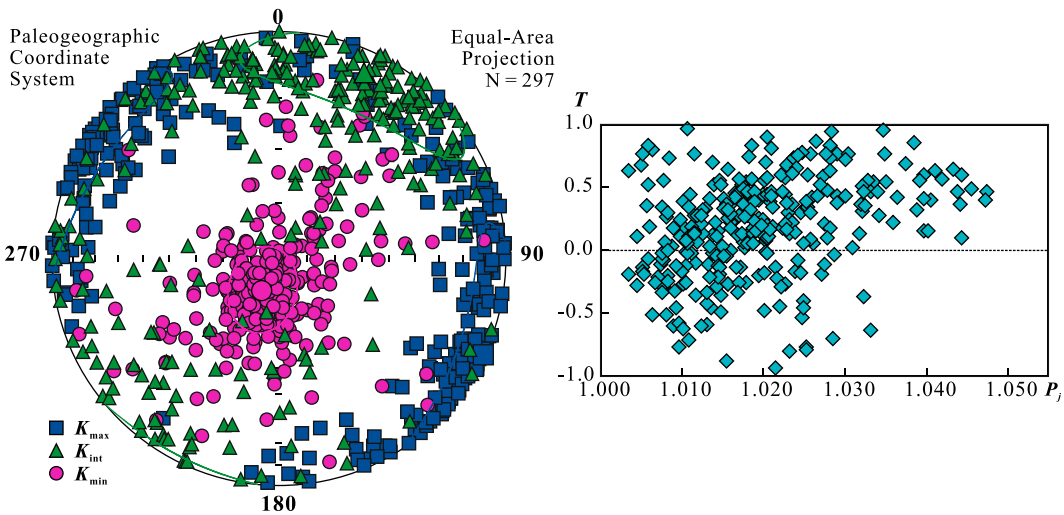



**Figure 5. Equal-area stereoplot of principal susceptibility axes [maximum ($K_{max}$), intermediate ($K_{int}$), and minimum ($K_{min}$)] in a stratigraphic coordinate system. Plots of anisotropy degree ($Pj$) versus shape parameter ($T$). N—number of samples.**



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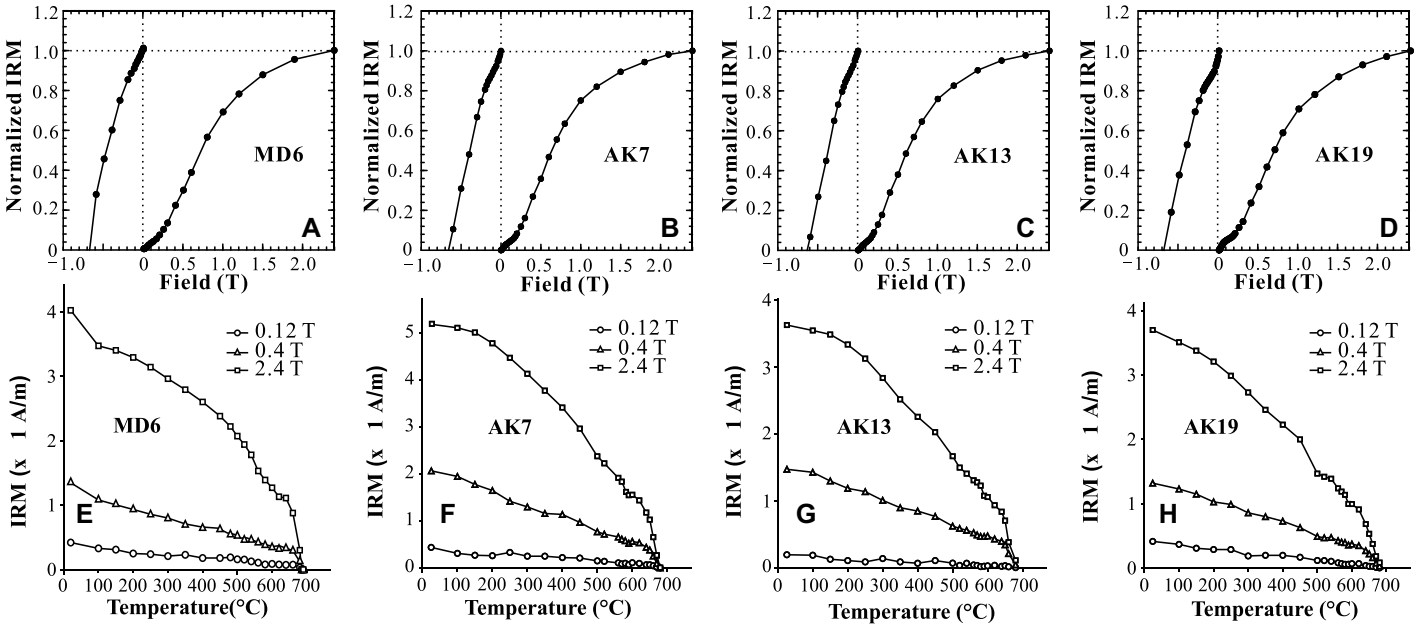



**Figure 6. Rock magnetic measurements for the representative samples from the Qiangtang Terrane, Tibetan Plateau. (A–D) Isothermal remanent magnetization (IRM) acquisition curves and backfield demagnetization of saturation IRM curves. (E–H) Thermal demagnetization of a three-component IRM following the method described by Lowrie (1990) for four representative specimens.**

ing that high-coercivity magnetic minerals are dominant. The backfield demagnetization of SIRM reduces to zero at 0.6–0.7 T. Thermal demagnetization of three-axis IRM shows that the soft, medium, and hard components eventually unblock ~680 °C, supporting that the high-coercivity magnetic mineral is hematite.

## Microscopic Observations

Microscopic observations show that the redbed specimens from the Madeng and Anduo areas contain abundant quartz, detrital hematite grains, and pigmentary hematite (Fig. 7). Detrital hematite grains appear as angular to subrounded with different sizes ranging from a few to tens of microns, whereas pigmentary hematite formed as the rims of clasts (Figs. 7C and 7F). These detrital hematite particles most probably record a detrital remanent magnetization (DRM). The random orientation of mineral particles supports that the studied redbeds did not suffer from significant deformation. These petrographic characteristics are similar with those observed from the Abushan Fm redbeds in the Shuanghu (Meng et al., 2018) and Linzhou (Ma et al., 2022) areas.

## Paleomagnetic Results

A total of 315 specimens were subjected to stepwise thermal demagnetization, and typical Zijderveld diagrams are illustrated in Figure 8A. Most specimens do not reveal a well-defined low unblocking temperature component close to the present geomagnetic field direction, but rather a well-defined high-temperature component (HTC) that decays to the origin, which is defined as the ChRM direction, and this can be isolated between 300–500 °C and 660–680 °C (Table S1[1]). Except for the two sites AK21 and AK22 drilled from the redbed cobbles, 262 specimens yield reliable ChRM directions that provide 37 site-mean directions. The eight paleomagnetic sites including 64 specimens collected from the southwest dipping limb of folds provide a mean direction (Group A) of $D_g = 21.5°$, $I_g = +1.8°$, $k_g = 60.3$, $\alpha_{95} = 7.2°$ in situ, and $D_s = 354.0°$, $I_s = +44.2°$, $k_s = 80.6$, $\alpha_{95} = 6.2°$ after tilt correction (Fig. 9A). Group A passes the McFadden (1990) fold tests at the 95% confidence level (Table 1). The 29 paleomagnetic sites including 198 specimens distributed at the north dipping limb of folds yield a mean direction (Group B) of $D_g = 122.9°$, $I_g = +71.8°$, $k_g = 14.3$, $\alpha_{95} = 7.3°$ in situ, and $D_s = 2.6°$, $I_s = +34.2°$, $k_s = 46.4$, $\alpha_{95} = 4.0°$ after tilt correction (Fig. 9B). Group B passes both McElhinny (1964) and McFadden (1990) fold tests at the 95% and 99% confidence levels (Table 1). The 37 paleomagnetic sites collected from the Abushan Fm redbeds yield a grand mean of $D_g = 74.6°$, $I_g = +68.3°$, $k_g = 4.0$, $\alpha_{95} = 13.6°$ in situ, and $D_s = 0.9°$, $I_s = +36.4°$, $k_s = 43.6$, $\alpha_{95} = 3.6°$ after tilt correction (Fig. 9C), corresponding to a Fisherian mean paleopole located at 78.2°N, 266.8°E with $A_{95} = 3.6°$ and a paleolatitude of 20.5° ± 3.6°N for the study area (32.3°N, 91.4°E). This overall-mean direction passes McElhinny (1964) and McFadden (1990) fold tests at the 99% confidence level (Table 1), and the Enkin (2003) direction-correction (DC) test is also positive with the best group at 88.90% ± 5.10% untilting (Fig. 9C).

The specimens drilled from redbed cobbles at the AK21 and AK22 sites show isolation of a stable HTC decaying toward the origin. The ChRM directions obtained from different cobbles at the same site are randomly distributed, while the two specimens collected from the same cobbles showed similar ChRM directions (Fig. 8B). The Watson's (1956) random distribution test for these two conglomerate sites yielded R values of 5.03 and 4.60, respectively, which were less than the $R_0$ value (n = 10) of 5.94 at the 99% confidence level and 5.03 at the 95% confidence level. Thus, positive fold and conglomerate tests strongly confirm that the Abushan Fm redbeds have recorded a primary remanent magnetization.

[1]Supplemental Material. The supplemental material contains characteristic remanent magnetization (ChRM) directions for the Abushan Formation redbeds (Table S1) and summary of Late Cretaceous redbeds paleomagnetic results from the Qiangtang Terrane (Table S2). Please visit https://doi.org/10.1130/GSAB.S.23638266 to access the supplemental material, and contact editing@geosociety.org with any questions.



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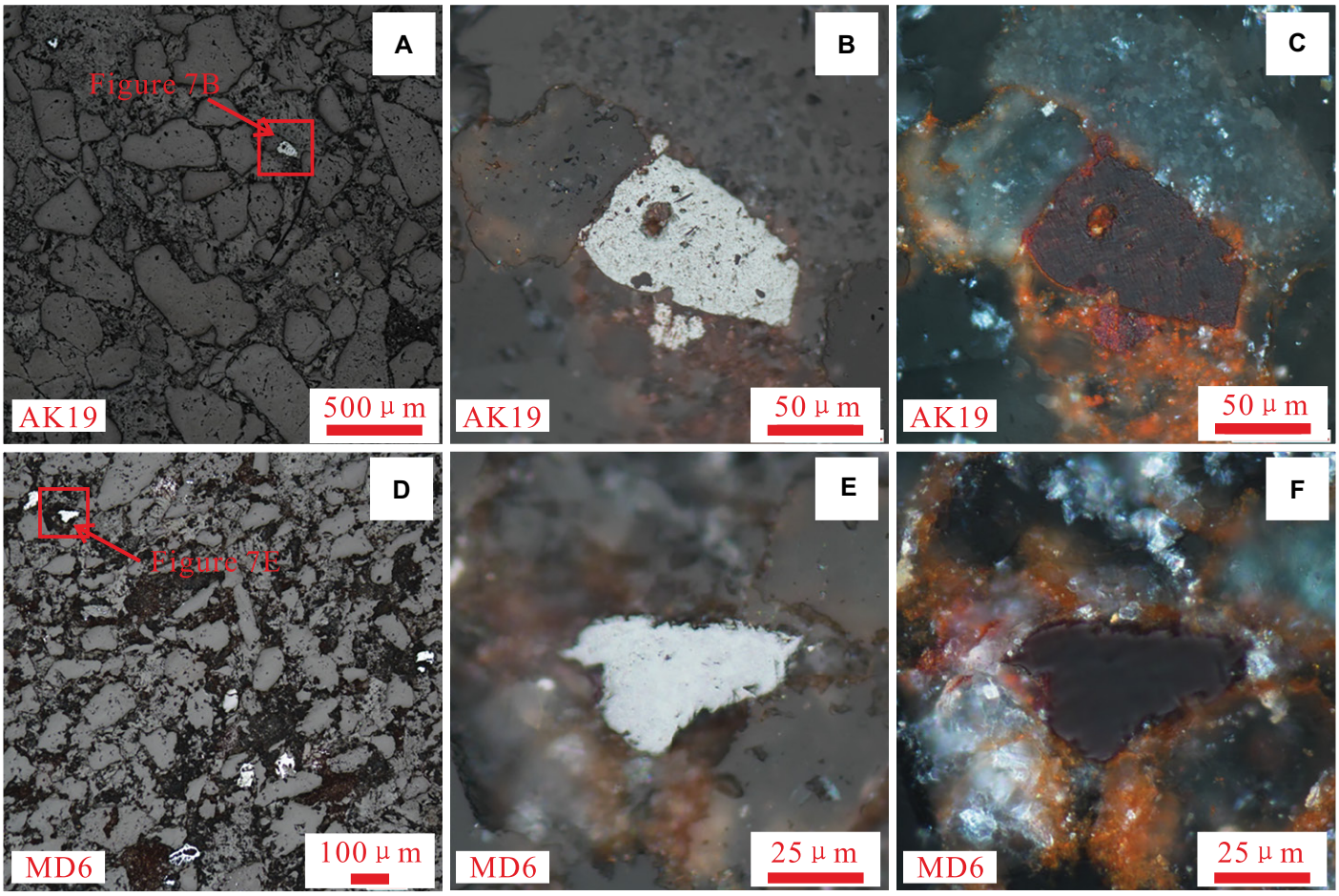



**Figure 7. Photomicrograph of the Abushan Formation redbeds of the Qiangtang Terrane, Tibetan Plateau. (A–D) Detrital hematite grains under plane-polarized reflected light. (E and F) Detrital hematite grains and surrounding pigmentary hematite under cross-polarized reflected light.**

## DISCUSSION

### The Reliability of the Abushan Fm Redbed Paleomagnetic Datasets

Though the positive fold and conglomerate tests indicate that the ChRM directions obtained from the Abushan Fm redbeds represent a primary magnetization, the mean inclination observed from the southwest dipping limb ($I_s = +44.2°$) is clearly steeper than that of the north dipping limb ($I_s = +34.2°$). The conspicuous inclination discrepancies in the Upper Cretaceous redbed paleomagnetic data may be explained by some factors such as: (1) insufficient averaging of geomagnetic paleosecular variation (e.g., Deenen et al., 2011; Vaes et al., 2021); (2) a secondary chemical remanent magnetization (CRM) recorded in the redbeds that partially or completely overprints the DRM (e.g., Kodama, 2012; Jiang et al., 2015, 2022); (3) the distortional strains should be taken seriously when the remanence directions are obtained from two limbs of folds (Cogné et al., 1991; Kanamatsu and Herrero-Bervera, 2006; Kligfield et al., 1983); (4) the depositional and/or compaction of redbeds may result in the ChRM inclination shallowing (e.g., Tauxe and Kent, 2004; Tan et al., 2010; Ding et al., 2015); and (5) the strata may not have been horizontal at the time of magnetization because of unidentified syntectonic growth strata (Gilder et al., 2001; Sato et al., 2011; Tong et al., 2019; Bian et al., 2020; Bian and Yang, 2021; Wang et al., 2022). These five factors are discussed separately to clarify the ChRM inclination discrepancy in the Abushan Fm redbeds obtained from two different strikes, as well as to assure the reliability of the paleomagnetic dataset we report.

#### *Tests for Paleosecular Variation*

Deenen et al. (2011) suggested the *N*-dependent $A_{95}$ envelope can help ascertain whether paleomagnetic datasets have sufficiently averaged out paleosecular variation. Considering that each independently oriented core has been drilled from different stratigraphic levels, we thus assume that each ChRM direction is a spot reading of the geomagnetic field, and thus that it can be transformed to a virtual geomagnetic pole (VGP). For the 37 paleomagnetic sites, the 262 VGPs yield a Fisherian specimen-mean paleopole at 78.6°N, 265.9°E with $A_{95} = 1.8°$ for the study area (32.3°N, 91.4°E). The $A_{95}$ value of 1.8° is within the *N*-dependent envelope ($A_{95min} = 1.3°$, $A_{95max} = 2.5°$) for 262 poles (Deenen et al., 2011). For the 64 specimens from the southwest dipping limb and 198 specimens from the north dipping limb, their corresponding VGPs provide the Fisherian specimen-mean paleopoles at 82.4°N, 313.7°E with $A_{95} = 3.5°$ and at 76.5°N, 258.0°E with $A_{95} = 2.0°$ for the study area (32.3°N, 91.4°E). These two $A_{95}$ values of 3.5° and 2.0° are also within their corresponding $A_{95}$ envelopes of ($A_{95min} = 2.3°$, $A_{95max} = 6.0°$) for 64 poles and ($A_{95min} = 1.4°$, $A_{95max} = 2.9°$) for 198 poles, respectively (Deenen et al., 2011). These tests support the inference that the above-mentioned three mean directions obtained from



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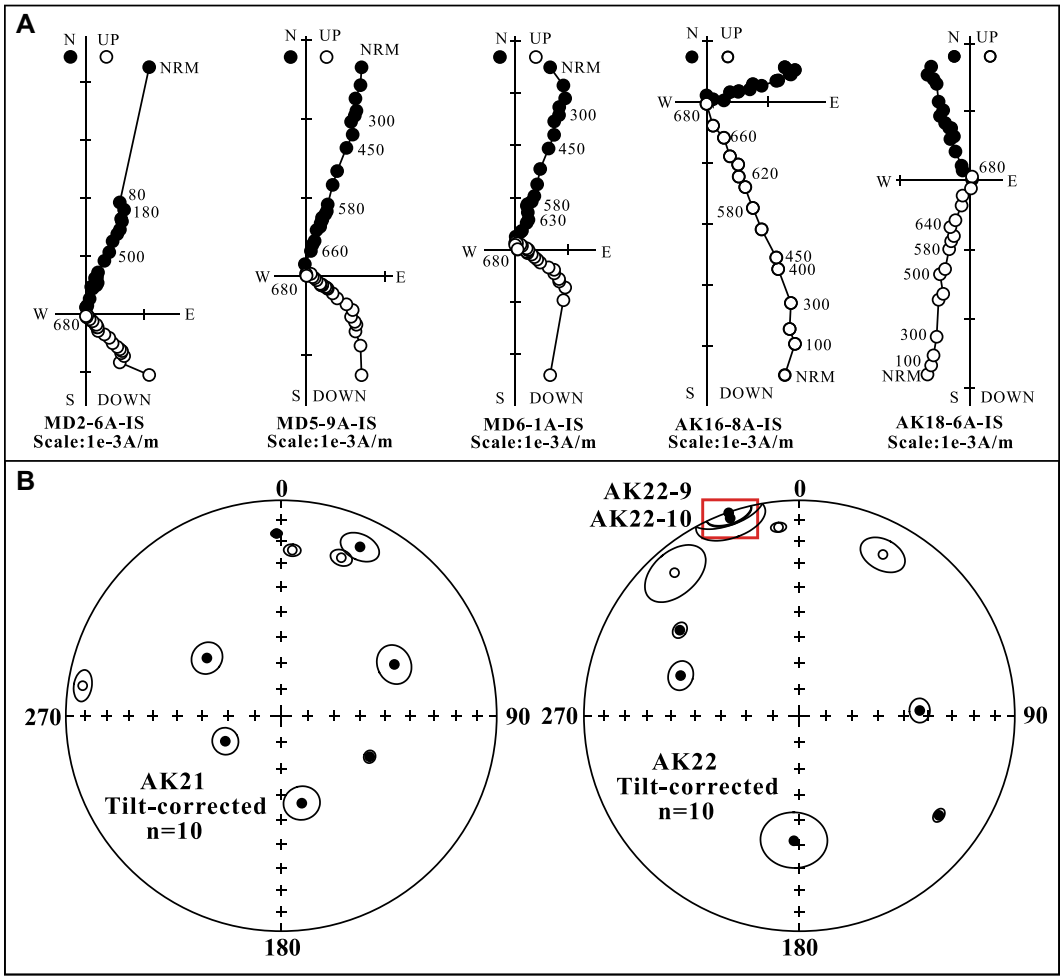



**Figure 8. (A) Demagnetization diagrams showing representative Abushan Formation redbeds of the Qiangtang Terrane, Tibetan Plateau, in geographic coordinates. The solid and open symbols represent the projections onto the horizontal and vertical planes, respectively. (B) Results of conglomerate test of the Abushan Formation redbeds. NRM—natural remanent magnetization; n—number of specimens.**

the Upper Cretaceous redbeds have sufficiently characterized geomagnetic secular variation.

### *Origin of Hematite*

The natural remanent magnetization in most redbeds is a mixture of detrital hematite and authigenic hematite (Tauxe et al., 1980). In general, a primary DRM is carried by detrital hematite, and it can partially to completely be replaced by a secondary CRM, which is recorded by authigenic hematite (Collinson, 1966; Tauxe et al., 1980; Tauxe and Kent, 1984; Huang et al., 2020); to identify the primary magnetization, an essential aspect of paleomagnetic studies on redbeds is to discriminate CRM from DRM (Walker et al., 1981; Kodama and Dekkers, 2004). Previous studies have suggested that the shapes of demagnetization curves and the maximum unblocking temperatures can discriminate between these two remanences (Jiang et al., 2015, 2022). The thermal demagnetization curves of the Abushan Fm redbeds show a convex up shape with the maximum unblocking temperature at ~680 °C (Fig. 10). Such demagnetization characteristics are clearly different from the concave shape of the CRM demagnetization curve, which has a wider laboratory unblocking temperature spectrum ~600–650 °C, but are fully consistent with the DRM demagnetization curve that displays a convex up shape with a confined narrow unblocking temperature up to 660–680 °C (Fig. 10). This result, combined with the positive conglomerate and fold tests, as well as the petrographic characteristics, shows that the ChRM directions obtained from the Abushan Fm redbeds is likely a primary DRM carried by detrital hematite.

### *The Effect of Distortional Strains*

Although previous studies have shown that the distortional strains can reorient the magnetic foliations and lineations (Weil and Yonkee, 2009; Hrouda and Chadima, 2020; Cao et al., 2022), it is still unclear whether the distortional strains affect paleomagnetic directions (Bian et al., 2020; Bian and Yang, 2021; Zhao et al., 2021). According to our AMS analysis, the magnetic fabrics observed should belong to the incipient deformation (stage 2) to pencil structure (stage 3) (Hrouda and Chadima, 2020). Noticeably, our overall-mean direction passes both the McElhinny (1964) and McFadden (1990) fold tests and the Enkin (2003) DC test, as well as the positive conglomerate tests, which strongly support that ChRM directions should be a pre-folding primary magnetization, not secondary remagnetization as a result of penetrative deformation.

In addition, several studies have shown that the primary magnetization can be maintained when the deformed magnetic fabrics of the redbeds display characteristics of deformation stage 2 (incipient deformation) to stage 4 (weak cleavage) (e.g., Li et al., 2020a, 2020b; Ma et al., 2022; Wang et al., 2022). Thus, the effect of the distortional strains on remanence directions of the Abushan Fm redbeds can be ignored.

### *Tests for Inclination Shallowing*

Although three available Cretaceous paleomagnetic investigations on basalt flows intercalated with redbeds demonstrated that the two sets of site-mean directions for redbeds and basalt flows are quite consistent (Sun et al., 2006; Li et al., 2013; Cao et al., 2017a), the depositional and/or compaction-induced inclination shallow-



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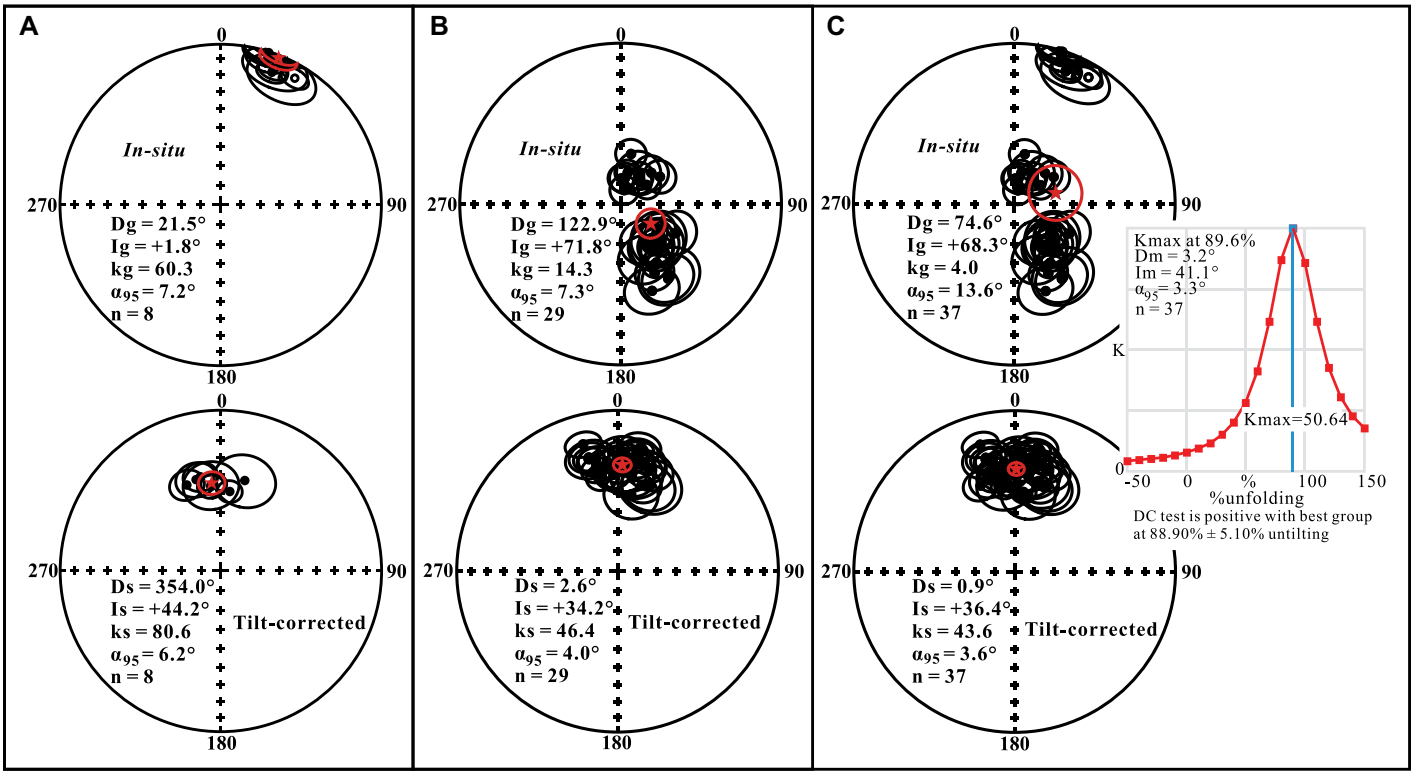



**Figure 9. Equal-area projections of estimated site-mean paleomagnetic directions from sites that yield interpretable results from the Abushan Formation redbeds of the Qiangtang Terrane, Tibetan Plateau. The stars indicate the overall-mean direction. Results of partial untilting test (Watson and Enkin, 1993) and direction-correction (DC) fold test (Enkin, 2003) of 37 accepted site-mean directions. Dg and Ig—declination and inclination in geographic coordinates; Ds and Is—declination and inclination in stratigraphic coordinates; kg and ks—precision parameter; $\alpha_{95}$—radius of the circle of 95% confidence; Dm and Im—declination and inclination at 89.6% unfolding; Kmax—precision parameter at 89.6% unfolding; n—number of sites.**

ing in redbeds is still an unresolved issue, which may result in inclination divergence (e.g., Tauxe and Kent, 2004; Tan et al., 2010; Kodama, 2012; Huang et al., 2013). Therefore, it is necessary to carry out inclination shallowing correction for the obtained Abushan Fm redbeds paleomagnetic datasets. Both the anisotropy-based correction method (Hodych and Buchan, 1994) and the E/I correction method (Tauxe and Kent, 2004) have been used widely to quantitatively test and correct for inclination shallowing of paleomagnetic data obtained from sedimentary rocks. Considering that the specimens collected from the north dipping limb show a shallower mean inclination than those from the southwest dipping limb, we therefore follow the abovementioned two methods to test and correct the paleomagnetic data obtained from the north dipping limb.

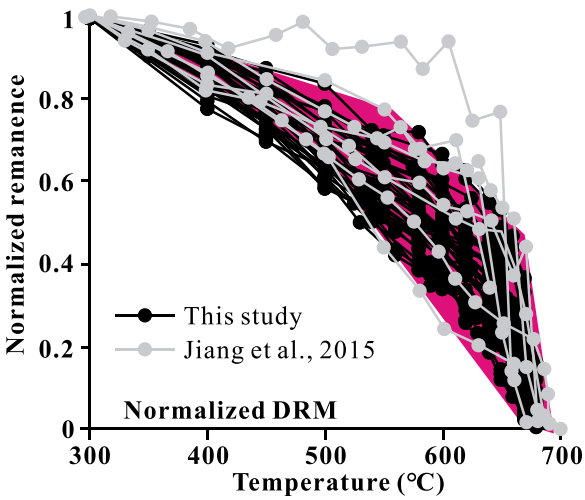



**Figure 10. Normalized thermal demagnetization curves of specimens of Abushan Formation redbeds of the Qiangtang Terrane, Tibetan Plateau, together with the reference data in Jiang et al. (2015). DRM—detrital remanent magnetization.**

The anisotropy-based correction results of the eight specimens from the north dipping limb are given in Figure 11. The values of the IRMz/IRMx in a field interval of 200–1000 mT and the thermal demagnetization temperature range of 600–680 °C were 0.900–1.017 and 0.903–1.021, respectively (Fig. 11). The IRMz/IRMx values of the eight specimens in the temperature range of 600–680 °C were used to calculate the mean IRMx/IRMz value (0.971) for correcting the measured overall-mean direction obtained from the north dipping limb. After the anisotropy-based inclination shallowing correction, the measured mean inclination of the 29 sites changed from 34.2 to 35.3, showing that the inclination shallowing is insignificant in the Abushan Fm redbeds.

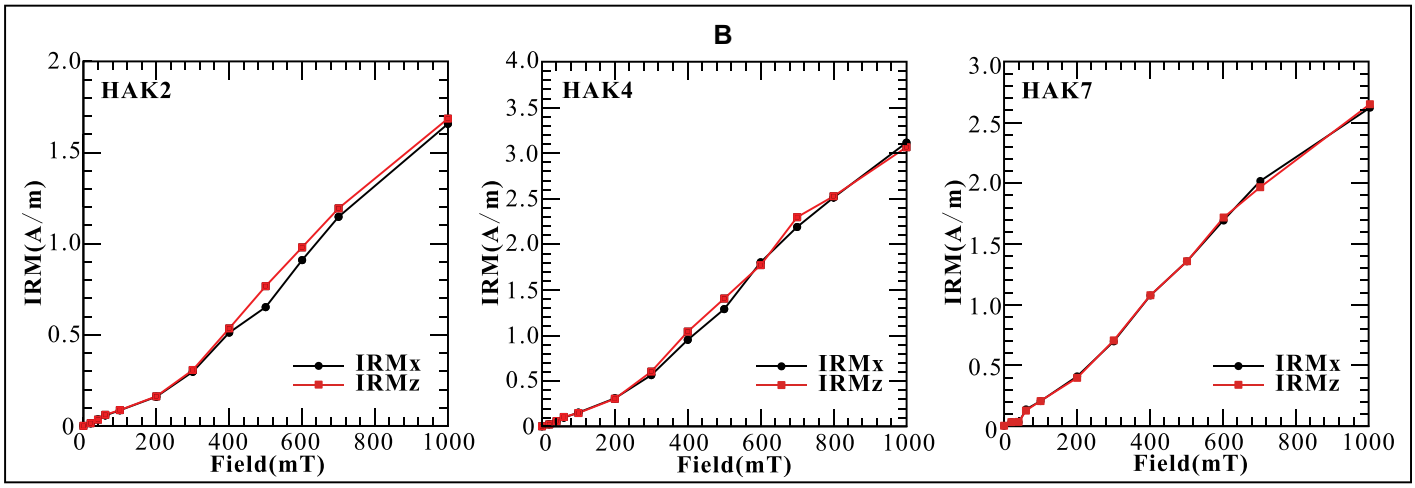
For three of the four sampling sections that were distributed at the north dipping limb (Fig. 1B; Table 1), only Section B contains more than 100 specimens and meets the minimum

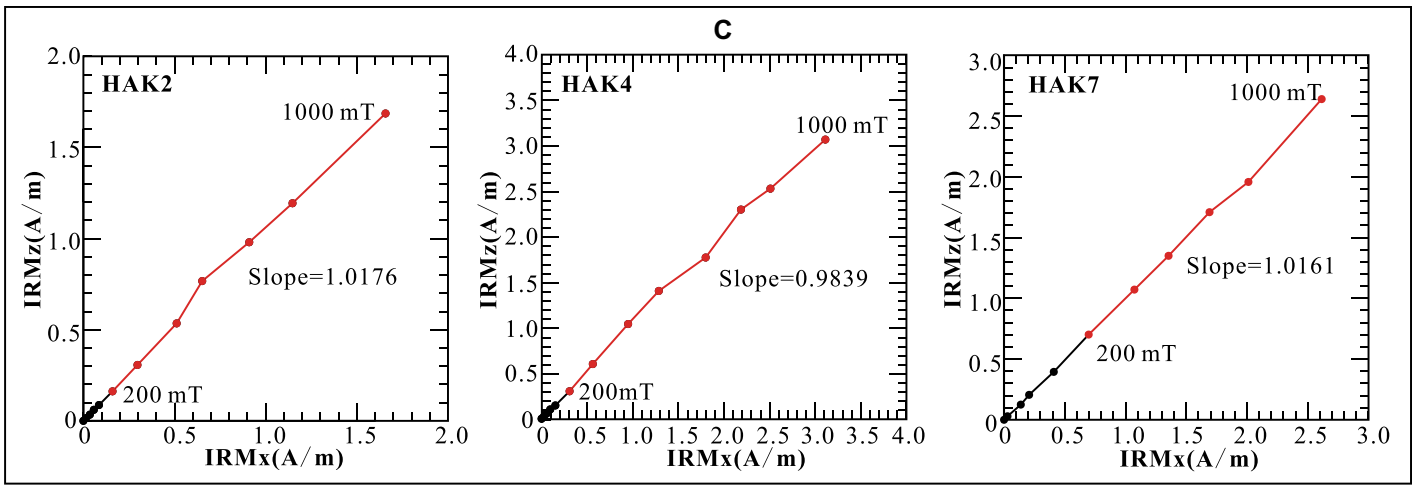


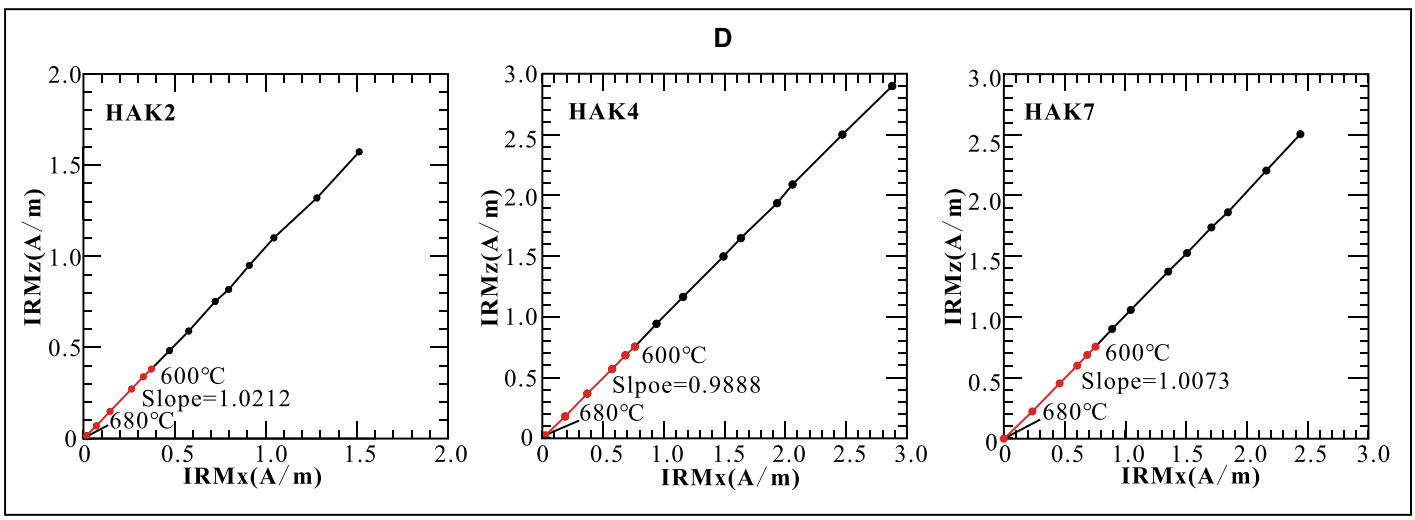
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A

| Sample Name | HAK1 | HAK2 | HAK3 | HAK4 | HAK5 | HAK6 | HAK7 | HAK8 | Mean |
|---|---|---|---|---|---|---|---|---|---|
| IRMz/IRMx (200-1000 mT) | 0.965 | 1.017 | 0.900 | 0.984 | 0.942 | 0.981 | 1.016 | 0.966 | 0.971±0.040 |
| IRMz/IRMx (600-680 °C) | 0.967 | 1.021 | 0.946 | 0.989 | 0.903 | 0.955 | 1.007 | 0.983 | 0.971±0.040 |
| Corrected overall-mean inclination | 35.4° | 33.5° | 36.2° | 34.6° | 37.9° | 35.8° | 34.0° | 34.8° | 35.3 ± 1.4° |






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**Figure 11. Results of anisotropy of isothermal remanent magnetization experiments. (A) Anisotropy of isothermal remanent magnetization of the Abushan Formation of the Qiangtang Terrane, Tibetan Plateau, and the corrected overall-mean inclination. (B) Plots of IRMx (parallel to bedding) and IRMz (perpendicular to bedding) acquisitions produced by applying a magnetic field at an angle of 45° to the bedding as a function of increasing field. (C) The slope (IRMz/IRMx) of the least-squares-fit for data points between 200 mT and 1000 mT was used as a measure of the magnetic anisotropy of hematite. (D) The slope of the thermal demagnetization of IRMz and IRMx between 600 °C and 680 °C was used as a measure of the inclination shallowing ratio for the inclination correction of the redbed samples.**

size requirement. The 124 specimens of Section B provide a tilt-corrected specimen-mean direction of $D_s = 10.7°$, $I_s = +34.6°$, $k_s = 30.2$, $\alpha_{95} = 2.4°$. Applying the E/I method to the 124 ChRM directions indicates an inclination flattening factor ($f$) of 0.98, and the corresponding E/I corrected inclination increases to 34.8° with an error range of 33.4°–41.1° (Fig. 12). Undoubtedly, the E/I shallowing corrected inclination of ~34.8° is fully consistent with the measured specimen-mean inclination of ~34.6° within the confidence levels, further supporting that the sampled Abushan Fm redbeds did not experience significant inclination shallowing. Therefore, the shallower mean inclination observed in the north dipping limb cannot be attributed to the depositional and/or compaction-induced inclination shallowing.

### *Syntectonic-Sedimentation Correction*

The Abushan Fm redbeds were deposited in an alluvial fan to braided river environment (Zhang et al., 1985; Ma et al., 2017a), which, combined with the fluvial gravel deposits observed in our sampling sections (Fig. 3B), supports the interpretation that the Abushan Fm redbeds are syntectonic growth strata and their original bedding might not be horizontal when the sediments were magnetized. Therefore, the conspicuous inclination discrepancy obtained from two different limbs of the folded Abushan Fm redbeds may be attributed to syntectonic-sedimentation (Fig. 13A).

The syntectonic-sedimentation-correction of Tong et al. (2019) suggests that the original dip of the sampling Abushan Fm redbeds is 7.5° (Fig. 13B). After the syntectonic-sedimentation-correction, the site-mean direction for the southwest-dip strata is $D_{c1} = 0.4°$, $I_{c1} = +41.2°$, $k_{c1} = 81.7$, $\alpha_{95} = 6.2°$, $n = 8$, and for the north-dip strata it is $D_{c2} = 5.3°$, $I_{c2} = +40.9°$, $k_{c2} = 53.3$, $\alpha_{95} = 3.7°$, $n = 29$ (Fig. 13D). The site-mean direction for the total 37 sites is $D_c = 4.2°$, $I_c = +41.0°$, $k_c = 57.7$, $\alpha_{95} = 3.1°$ (Fig. 13D), corresponding to a paleopole at 80.4°N, 247.5°E with dp/dm = 2.3/3.8 and a paleolatitude of 23.5° ± 2.3°N for the study area (32.3°N, 91.4°E). This syntectonic-sedimentation-correction overall-mean direction passes both McElhinny (1964) and McFadden (1990) fold tests at the 95% and 99% confidence levels (Table 1), and the Enkin (2003) direction-correction (DC) test is also positive with the best group at 101.66% ± 7.79% untilting (Fig. 13C). Obviously, the site-mean inclination of the southwest dipping strata ($I_{c1} = 41.2°$) is similar to that of the north dipping strata ($I_{c2} = 40.9°$) after syntectonic-sedimentation-correction and the $k$ values increase significantly from 43.6 to 57.7, indicating that the inclination discrepancy obtained from Group A and Group B samples should result from the syntectonic sedimentation. In addition, considering that the overall-mean inclination ($I_c = 41.0° \pm 3.1°$) after syntectonic-sedimentation-correction is also relatively consistent with the observed overall-mean inclination ($I_s = 36.4° \pm 3.6°$), the paleomagnetic data before correction are still credible for paleogeographic reconstructions. Several paleomagnetic datasets of the Cretaceous redbeds obtained from two limbs of folds also show that the inclinations are generally consistent before and after syntectonic-sedimentation-correction (e.g., Bian et al., 2020; Wang et al., 2022).

The abovementioned studies demonstrate that the conspicuous inclination discrepancies in the Upper Cretaceous redbeds of the Lhasa and Qiangtang terranes mainly result from the lack of correction for syntectonic growth processes. Although the syntectonic-sedimentation can cause a significant deviation of paleomagnetic inclinations obtained from monoclinal strata, paleomagnetic datasets observed from two limbs of folds are still reliable. In addition, previous paleomagnetic results obtained from the Lhasa and Qiangtang terranes indicated that inclination shallowing is negligible within the

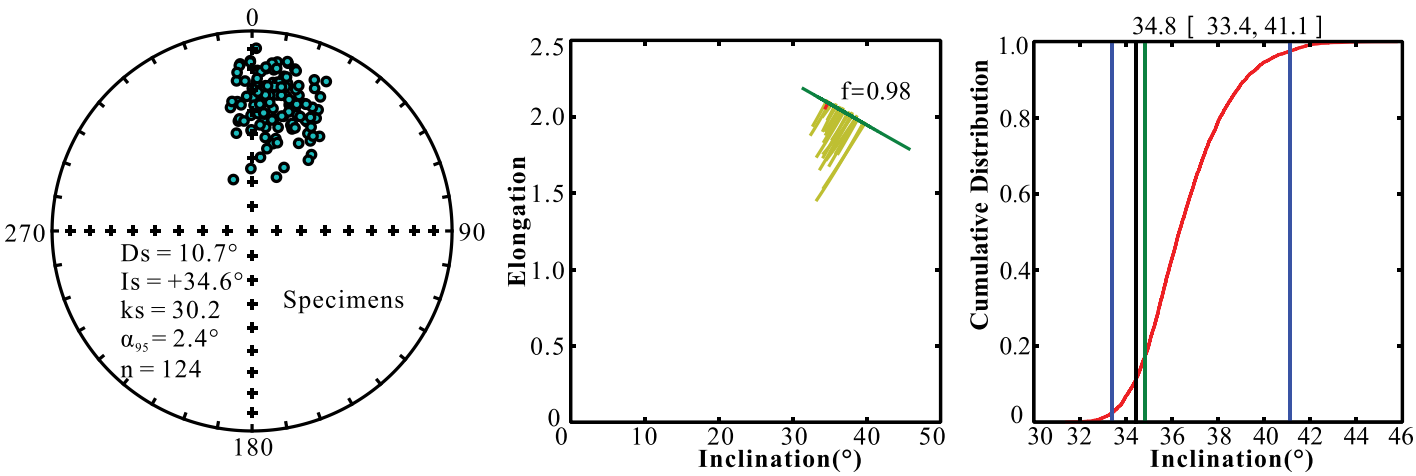



**Figure 12. The elongation/inclination correction (Tauxe and Kent, 2004) results for the high-temperature component of Section B (Qiangtang Terrane, Tibetan Plateau). From left to right are the equal-area projection of specimen-mean directions, elongation direction of the curve with respect to inclination, inclination flattening factor ($f$), and cumulative distribution of the corrected inclination. Ds and Is—declination and inclination in stratigraphic coordinates; ks—precision parameter; $\alpha_{95}$—the radius that the mean direction lies within 95% confidence; n—number of specimens.**



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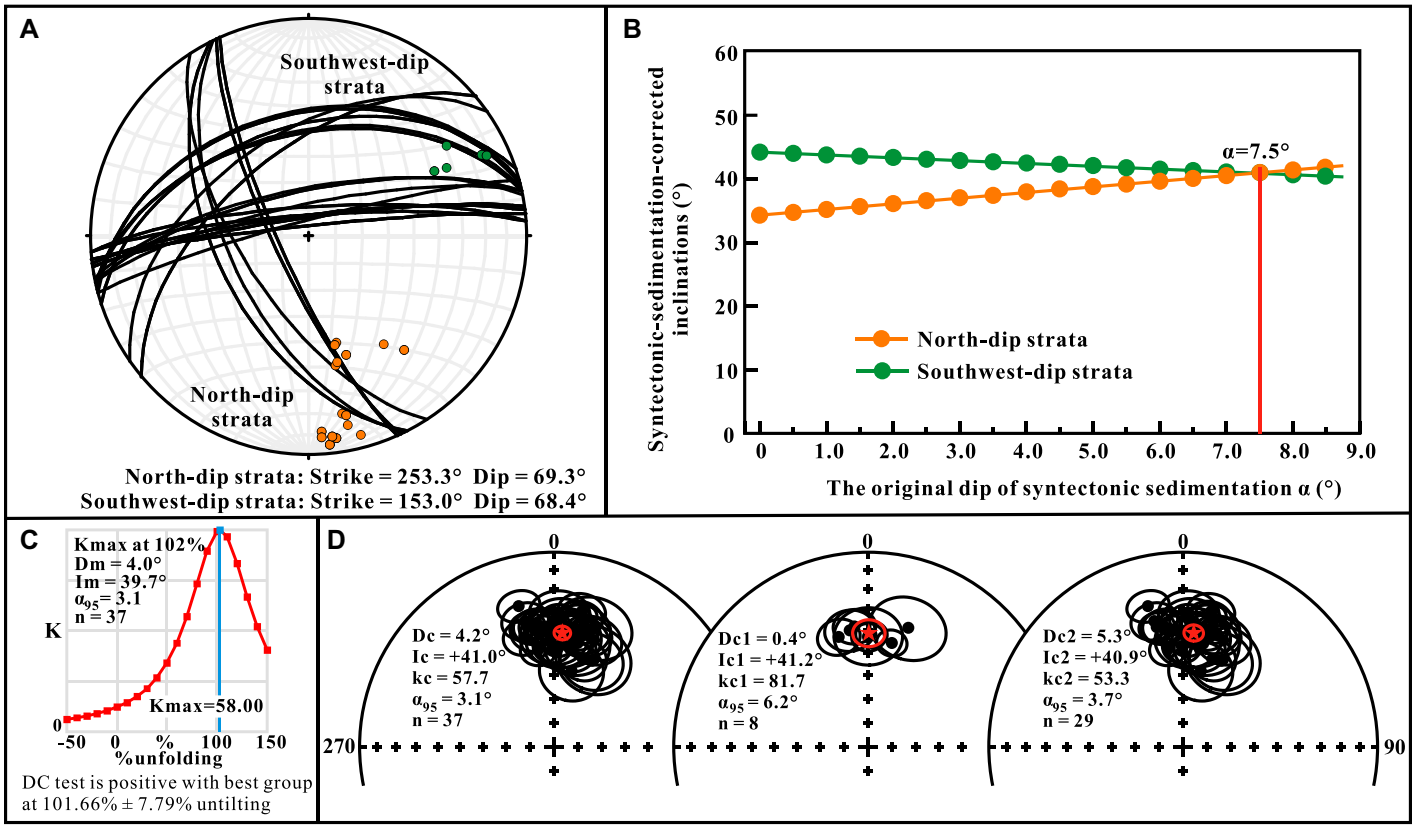



**Figure 13. (A) Lower hemisphere stereonet projection of bedding attitudes of the Abushan Formation (Fm) redbeds of the Qiangtang Terrane, Tibetan Plateau. (B) The syntectonic-sedimentation-correction for the Abushan Fm redbeds. (C) Results of partial untilting test (Watson and Enkin, 1993) and direction-correction (DC) fold test (Enkin, 2003) of 37 site-mean directions after syntectonic-sedimentation-correction. (D) Equal-area projections of site-mean directions for syntectonic-sedimentation-corrected paleomagnetic data of the Abushan Fm redbeds. Dc and Ic—declination and inclination after syntectonic-sedimentation-correction; kc—precision parameter; $\alpha_{95}$—radius of the circle of 95% confidence; Dm and Im—declination and inclination at 102% unfolding; Kmax—precision parameter at 102% unfolding; n—number of sites**

Upper Cretaceous redbeds. However, most of the Cretaceous paleomagnetic datasets from the Lhasa terrane were obtained from either the single north dipping limb of folds (e.g., Tan et al., 2010; Tong et al., 2017; Li et al., 2017a) or the single south dipping limb of folds (e.g., Chen et al., 2012). Therefore, the unidentified Cretaceous syntectonic growth strata would lead to the shallower mean inclination observed in the single north dipping limb and the steeper mean inclination observed in the single south dipping limb.

TABLE 2. SUMMARY OF THE LATE CRETACEOUS PALEOMAGNETIC RESULTS FROM THE QIANGTANG TERRANE, TIBETAN PLATEAU

| ID | Area | Location | | Age (Ma) | Paleopole | | | Paleolat (°N) | Latitude difference (°) | n/N | Criterion(R) | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $S_{lat}$ (°N) | $S_{lon}$ (°E) | | Plat (°N) | Plon (°E) | $A_{95}$(dp/dm) (°) | | | | | |
| West of the EHS | | | | | | | | | | | | |
| GZ | Gaize | 32.5 | 84.3 | K2–103.8 | 45.4 | 348.1 | 3.1 | 18.6 ± 3.1 | 13.9 ± 3.1 | 174/22 | 123F5X7(6) | Chen et al. (2017) |
| SH | Shuanghu | 33.2 | 88.7 | 97.0 ± 14.0 | 29.2 | 171.8 | 2.5/4.3 | 20.8 ± 2.5 | 12.4 ± 2.5 | 71/9 | 123F5X7(6) | Meng et al. (2018) |
| **AD** | **Anduo** | **32.3** | **91.4** | **~83.6–89.1** | **77.9** | **267.9** | **2.4/4.2** | **20.2 ± 2.4** | **12.1 ± 2.4** | **264/37** | **123F5X7(6)** | **This study** |
| East of the EHS | | | | | | | | | | | | |
| MK1 | Mangkang | 29.7 | 98.4 | K2 | 47.0 | 165.1 | 6.7/9.3 | 36.6 ± 6.7 | | 186/17 | 123F5R7(7) | Tong et al. (2015) |
| MK2 | Mangkang | 29.7 | 98.7 | K2 | 56.7 | 172.7 | 10.6 | 32.9 ± 10.6 | | 59/11 | 123F5X7(6) | Huang et al. (1992) |
| **Mean paleolatitude obtained using the inclination-only mean arithmetic method (reference point: 29.7°N, 98.5°E)** | | | | | | | | **33.9 ± 4.3** | **–4.2 ± 4.3** | | | **This study** |

*Note:* $S_{lat}$—latitude of sites; $S_{lon}$—longitude of sites; K2—Late Cretaceous; Plat—latitude of poles; Plon—longitude of poles; $A_{95}$—the radius that the mean pole lies within 95% confidence; dp/dm—semi-axes of elliptical error of the pole at a probability of 95%; Paleolat—paleolatitude calculated for the reference point located at the studied area; Latitude difference—difference between the sampling latitude and the corresponding paleolatitude; n/N—number of samples/sites used to calculate Fisher mean; Criterion(R)—data quality criteria (number of criteria met) after Meert et al. (2020); 1—well-determined rock age and a presumption that magnetization is the same age; 2—techniques and statistical analysis; 3—evaluation of remanence carriers; 4—field tests that constrain age of magnetization; 5—structure control and tectonic coherence with craton or block involved; 6—the presence of reversals; 7—no resemblance to paleopoles of younger ages [by more than a period]; F—positive fold test; R—positive reversal test; X—failed to meet this criterion. The data calculated in this study are bold.



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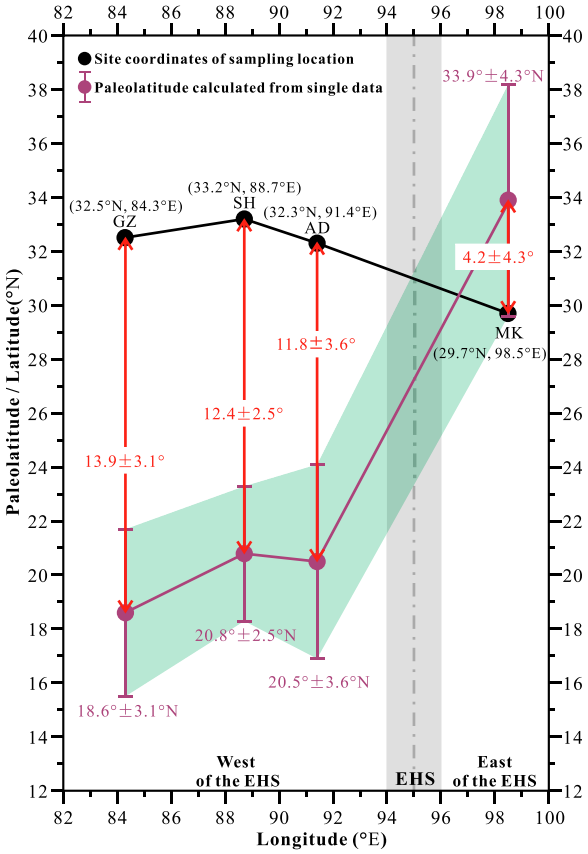



**Figure 14. Latitudinal differences between the individual sampling localities (Qiangtang Terrane, Tibetan Plateau) latitudes and the paleolatitudes during the Late Cretaceous. EHS—Eastern Himalaya Syntaxis. See Table 2 for the values, abbreviations, and references of the paleopoles AD, GZ, MK, and SH.**

## The Shape of the Qiangtang Terrane during the Late Cretaceous

High-quality paleomagnetic datasets are pre-condition for quantitative paleogeographic reconstruction. Meert et al. (2020) updated the Van der Voo (1990) criteria and proposed the "R" reliability indexes, which are listed in the notes of Table 2, to evaluate the quality of paleomagnetic data. Considering the age range of the Cretaceous Normal Polarity Superchron (ca. 121.4–83.6 Ma; Ogg, 2020) and the insignificant inclination shallowing in the central Tibetan Plateau, we require that reliable paleomagnetic data should fulfill at least five "R" indexes (1–4 and 7) with at least a positive field test.

Table 2 summarizes the five available Late Cretaceous paleopoles from the QT, and their paleolatitudes have been calculated for the individual sampling location. The MK1 paleopole satisfies all seven "R" indexes, and the other four paleopoles (GZ, SH, AD, and MK2) fulfill six "R" indexes (except for criteria 6, the presence of reversals). Therefore, all five datasets from redbeds match our selection criterion (Table 2). Three paleomagnetic poles (GZ, SH, and AD), which were obtained from the west of the EHS (Fig. 1A), yield paleolatitudes of 18.6° ± 3.1°N (Chen et al., 2017) for the Gaize area (32.5°N, 84.3°E), 20.8° ± 2.5°N (Meng et al., 2018) for the Abushan Fm of the Shuanghu area (33.2°N, 88.7°E), and 20.2° ± 2.4°N (this study) for the Abushan Fm of the Anduo area (32.3°N, 91.4°E), respectively. Comparing these three observed paleolatitudes with their sampling latitudes gives the latitudinal differences of 13.9° ± 3.1° (GZ), 12.4° ± 2.5° (SH), and 11.8° ± 3.6° (AD), indicating that the Gaize, Shuanghu, and Anduo sampling areas moved northward 1540 ± 340 km, 1380 ± 280 km, and 1310 ± 400 km after the Late Cretaceous, respectively (Fig. 14). Furthermore, the consistently northward displacement distance is within the paleomagnetic resolution for these three sampling locations, which span a wide longitude ranging from 84.3°E to 91.4°E, which supports that the QT shape to the west of the EHS was essentially similar to its present-day relatively E-W alignment during the Late Cretaceous.

The other two poles (MK1 and MK2) are from the Mangkang area, located to the east of the EHS, and provide paleolatitudes of 36.6° ± 6.7°N (Tong et al., 2015) for the Zonggu Fm in the MK1 sampling location (29.7°N, 98.4°E) and 32.9° ± 10.6°N (MK2; Huang et al., 1992) for the Mangkang Fm in the MK2 sampling locations (29.7°N, 98.7°E). Considering that these two redbed poles from the same sampling area have been obtained from the same age rocks and lithology, we follow the method of Arason and Levi (2010) to obtain the inclination-only (corresponding to paleolatitude-only) mean and then calculate its corresponding paleolatitude. A total of 28 paleomagnetic sites yield an inclination-only mean of 53.3° ± 4.5°, corresponding to a paleolatitude of 33.9° ± 4.3°N (Table S2; see footnote 1). Comparing this observed latitude with the sampling latitude of 29.7°N shows the latitude differences of 4.2° ± 4.3°, implying that the Mangkang sampling area to the east of the EHS moved southward 470 ± 480 km after the Late Cretaceous (Fig. 14). The opposite displacements sense between the west and east areas of the EHS has led to the QT shape around the EHS changing from a Late Cretaceous NE-SW alignment to its approximate present-day SE-NW alignment. These paleomagnetic observations are consistent with the previous geologic studies that indicate that the southeastern edge of the Tibetan Plateau, such as the Indochina and Shan-Thai block, has experienced a southward displacement from 700 ± 200 km to more than 1000 km caused by the indentation of India into Asia (e.g., Yang and Besse, 1993; Schärer et al., 1994; Yang et al., 1995; Leloup et al., 1995, 2001; Lacassin et al., 1996, 1997; Sato et al., 1999, 2007; Otofuji et al., 2007; Li et al., 2017a).

Due to the limited paleomagnetic data obtained from the Late Cretaceous rocks of the QT, as well as the discrepancy in data quality, further paleomagnetic studies from the Late Cretaceous rocks are still necessary to accurately constrain the shape of the QT prior to the India-Asia collision.

## CONCLUSIONS

We have conducted a combined geochronologic and paleomagnetic study on the Upper Cretaceous Abushan Fm redbeds of the Anduo area in the QT. These new paleomagnetic data pass both positive fold and conglomerate tests, and provide an opportunity to evaluate the reliability of redbed paleomagnetic datasets from the Tibetan Plateau. Based on the paleomagnetic data from the Upper Cretaceous redbeds of the QT, the main conclusions can be summarized as follows:

(1) The Anduo area of the QT was located at 20.2° ± 2.4°N during the Late Cretaceous.

(2) The reliability of redbed paleomagnetic datasets is not influenced by the distortional strains in the pencil structure magnetic fabrics stage.

(3) The conspicuous inclination discrepancies observed from the folded Upper Cretaceous redbeds of the Lhasa and Qiangtang terranes likely result from an inability to recognize syntectonic growth strata.



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(4) Although syntectonic-sedimentation may cause a significant deviation of paleomagnetic inclinations obtained from monoclinal strata, paleomagnetic datasets observed from two limbs of folds are still reliable.

(5) The QT shape to the west of the EHS was similar to its present-day relatively E-W alignment, and around the EHS its shape changed from the Late Cretaceous NE-SW alignment to its approximate present-day NW-SE alignment.

(6) The reliable Late Cretaceous paleomagnetic results from the QT indicate that the west of the EHS underwent a northward movement while the east of the EHS experienced a very slight southward movement after the Late Cretaceous.

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